Random Matrices and L-Functions

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Abstract

This thesis explores the profound connections between random matrix theory and number theory, particularly focusing on the statistical distribution of zeros of L-functions. We begin by introducing the Gaussian Unitary Ensemble (GUE) and the classical compact groups, detailing the eigenvalue statistics that serve as models for number-theoretic objects. The core of the work examines the Montgomery-Odlyzko law, which reveals that the pair correlation of the non-trivial zeros of the Riemann zeta function coincides with that of the GUE. We then discuss the Katz-Sarnak philosophy, which extends this connection to families of L-functions, showing that the distribution of low-lying zeros corresponds to the eigenvalue distribution near 1 of a classical compact group, with the symmetry type being a fundamental characteristic of the family. A detailed case study on the 1-level density for families of quadratic Dirichlet L-functions is presented to illustrate these concepts. Finally, we survey the modern Langlandsian framework for families, linking symmetry types to the representation theory of L-groups.

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1 Introduction

"This is the kind of mathematics that I enjoy, with many conjectures and few proofs. Deep mysteries remain, and the best is still to come". (F. Dyson)

The theory of random matrices originated in the late 1950s with the physicist Eugene Wigner, who introduced them to model the spectra of heavy atoms composed of many particles with unknown interactions. Unable to explicitly calculate the behavior of these systems, the idea arose to forgo them and provide a completely random description by considering sets of matrices, known as ensembles, with randomly distributed entries obeying certain symmetry constraints. Recently, random matrix theory has permeated various areas of mathematics and physics, including number theory, operator theory, random permutations, and random growth models [**akemann2011oxford**]. This work addresses the relationship between random matrix theory and number theory.

One of the most emblematic unsolved problems in mathematics is the Riemann Hypothesis. Implicitly, this conjecture states that the non-trivial zeros of the Riemann zeta function lie on the line $\mathrm{Re}(s) = 1/2$ of the complex plane. Around 1927, Hilbert and Pólya suggested that there might

be a spectral interpretation of the zeros of the Riemann zeta function. That is, they conjectured that it might be possible to establish an explicit correspondence between the non-trivial zeros of the Riemann zeta function and the eigenvalues of some operator acting on a Hilbert space.

The first significant evidence for the Hilbert-Pólya conjecture came in 1972 when the mathematician Hugh Montgomery [**montgomery1973pair**] calculated the pair correlation of the Riemann zeta function. He found that its zeros exhibit the same repulsion as the eigenvalues of large unitary matrices (i.e., as the matrix size tends to infinity), a fact pointed out by the physicist Freeman Dyson. Montgomery conjectured more general analogies related to random matrices, which were confirmed numerically by Andrew Odlyzko in the 1980s. This phenomenon is known as the Montgomery-Odlyzko law.

Furthermore, in 1995, Rudnick and Sarnak [**rudnick1996zeros**] determined the $n \geq 2$ correlations for the zeros of the Riemann zeta function, as well as for more general L-functions. These results are in perfect agreement with predictions from random matrix theory. Unfortunately, the universal behavior described by the Montgomery-Odlyzko law does not provide information about what the relevant specific operator should be for each L-function.

The philosophy of Katz and Sarnak [**katz1999random**] teaches us that the situation is, in fact, even richer when studying the distribution of zeros of families of L-functions over function fields. Katz and Sarnak showed that, in function fields, the Montgomery-Odlyzko law holds true, but also that the laws associated with the distribution of zeros near the critical point are distinct. These laws correspond to eigenvalues of random matrices from the classical compact groups (unitary, orthogonal, and symplectic). In this case, this behavior can be explained naturally in terms of the geometric monodromy group associated with each family. This is the type of result we aim to explain here. The only thing we can say in advance is that random matrix theory turns out to be a remarkable and coherent probabilistic model for describing the zeros of L-functions.

The structure and choice of topics in this text are based on [**katz1999zeros**] and aim to be an extension of some of the themes discussed there. The structure of this work is as follows.

In Chapter 1, we introduce a particular type of random matrix, the Gaussian Unitary Ensemble. We study the statistics associated with the eigenvalues of matrices in this ensemble. Of particular importance for subsequent chapters will be Dyson's theorem and the distribution of gaps between eigenvalues. A complete proof of the gap distribution formulas is provided.

In Chapter 2, we study other types of random matrices, the classical compact groups. We study the associated eigenvalue statistics: the n-level correlation, the distribution of gaps between eigenvalues, the n-level density, and the distribution of the first eigenvalue.

In Chapter 3, we examine Montgomery's article [**montgomery1973pair**], giving a complete detailed proof of the pair correlation via the dual theorem, thus motivating the Montgomery-Odlyzko law.

In Chapter 4, we discuss the generalization by Rudnick and Sarnak of Montgomery's theorem. The generalization occurs in two directions: first, the functions studied are extended, and then the results are extended to all correlations between zeros of L-functions. The second part of the chapter discusses the Katz-Sarnak philosophy, which proposes studying the zeros of families of L-functions located in neighborhoods around the critical point.

In Chapter 5, we present a detailed case study of the 1-level density for families of quadratic Dirichlet L-functions, demonstrating the symplectic symmetry of this family as per the Katz-Sarnak conjectures.

Finally, in Chapter 6, we provide a modern survey of the Katz-Sarnak program within the Langlands program, detailing the proof mechanisms—explicit formula, trace formulas, monodromy—and explaining how symmetry types are encoded in L-group representations via the Frobenius-Schur indicator. We give a detailed exposition of the Sarnak-Shin-Templier framework, including the

structural invariants, the classification of symmetry types, and the proof strategy for the density conjecture.

Notation

Given a complex number $s = \sigma + it$, we denote by $\mathrm{Re}(s)$ (resp. $\mathrm{Im}(s)$) the real part (resp. imaginary part) of s.

We adopt Landau's notation. The statement

$$f(x) = O(g(x)), \quad x \to \infty$$

is read as "f is big-O of $g(x)$". It is equivalent to

$$f(x) \ll g(x), \quad x \to \infty,$$

which is read as "f is less than less than $g(x)$". Both statements mean that there exists a constant $C > 0$ such that for sufficiently large x, $|f(x)| \leq Cg(x)$. The number C is called the implied constant.

The statement

$$f(x) = o(g(x)), \quad x \to \infty$$

is read as "f is little-o of $g(x)$". Explicitly, it means

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0.$$

Equivalently, for every $C > 0$, if x is sufficiently large, $|f(x)| \leq Cg(x)$.

The statement $f(x) \sim g(x)$, read as "f is asymptotic to $g(x)$", means that $f(x) = (1 + o(1))g(x)$.

The Von Mangoldt function $\Lambda(n)$ is defined by

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p, k > 0 \\ 0 & \text{otherwise} \end{cases}$$

The prime counting function is

$$\pi(x) = \#\{p \leq x : p \text{ is prime}\}$$

For $x > 0$, the Chebyshev ψ function is defined as

$$\psi(x) = \sum_{n\leq x} \Lambda(n).$$

The Möbius function is denoted by

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n = p_1 p_2 \cdots p_k \\ 0 & p^2 \mid n. \end{cases}$$

We say a function $f : \mathbb{R}^n \to \mathbb{C}$ is of Schwartz class if f is smooth and both f and all its derivatives are rapidly decaying:

$$f^{(j)} \ll_{j,N} (1 + |x|)^{-A},$$

for every $A \geq 0$ and every integer $j \geq 0$.

Furthermore, if f is a Schwartz function, the Fourier transform of f is the function

$$\hat{f}(y) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x y} \, dx.$$

For Schwartz functions f and g, Plancherel's identity holds:

$$\int_{\mathbb{R}^n} \hat{f}(x) g(x) \, dx = \int_{\mathbb{R}^n} f(y) \hat{g}(y) \, dy.$$

2 Random Matrices: The Gaussian Unitary Ensemble (GUE)

2.1 The Space of Hermitian Matrices

Definition 2.1. *Let* $\mathrm{Herm}_N = \{H \in M_N(\mathbb{C}) : H = H^*\}$ *be the space of* $N \times N$ *Hermitian matrices.*

A matrix $H = (h_{ij})$ is Hermitian if $h_{ii} \in \mathbb{R}$ for all i, and $h_{ji} = \overline{h_{ij}}$ for all i, j.

Proposition 2.2. *The real dimension of* Herm_N *is* N^2.

Proof. The diagonal entries contribute N real parameters. For each of the $\binom{N}{2} = N(N-1)/2$ pairs $i < j$, the entry h_{ij} is a complex number, contributing 2 real parameters. Since h_{ji} is determined by h_{ij}, no further degrees of freedom are added. Thus the total real dimension is $N + 2 \cdot N(N-1)/2 = N^2$. □

2.2 Definition of the GUE

Definition 2.3. *A random matrix* $H = (h_{ij})_{1 \leq i,j \leq N}$ *belongs to the Gaussian Unitary Ensemble (*$\mathrm{GUE}(N)$*) if* $H \in \mathrm{Herm}_N$ *and its entries are independent (up to the Hermitian condition) Gaussian random variables with the following distributions:*

- *The diagonal entries* h_{ii} *are i.i.d. real Gaussian variables with mean* 0 *and variance* 1*:* $h_{ii} \sim \mathcal{N}(0, 1)$.
- *For* $i < j$*, the real and imaginary parts are i.i.d. real Gaussian variables with mean* 0 *and variance* 1/2*:*
$$\Re(h_{ij}) \sim \mathcal{N}(0, 1/2), \quad \Im(h_{ij}) \sim \mathcal{N}(0, 1/2).$$
- $h_{ji} = \overline{h_{ij}}$.

The variances are chosen so that $\mathbb{E}[|h_{ij}|^2] = 1$ for all i, j. The joint probability density function (with respect to the Lebesgue measure on $\mathrm{Herm}_N \cong \mathbb{R}^{N^2}$) is therefore the product of the individual densities:

$$f(H) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}} e^{-h_{ii}^2/2} \prod_{1 \leq i < j \leq N} \frac{1}{\sqrt{\pi}} e^{-2(\Re(h_{ij})^2 + \Im(h_{ij})^2)}.$$

2.3 Density of the GUE

Proposition 2.4. *The density can be written compactly as*

$$f(H) = C_N^{-1} e^{-\frac{1}{2}\operatorname{Tr}(H^2)},$$

where C_N is a normalization constant.

Proof. We compute $\operatorname{Tr}(H^2)$. For a Hermitian matrix, $h_{ij}h_{ji} = |h_{ij}|^2$. Thus

$$\operatorname{Tr}(H^2) = \sum_{i,j} h_{ij}h_{ji} = \sum_i h_{ii}^2 + 2\sum_{i<j} |h_{ij}|^2.$$

The exponent in $f(H)$ is

$$-\frac{1}{2}\sum_i h_{ii}^2 - \sum_{i<j} |h_{ij}|^2 = -\frac{1}{2}\left(\sum_i h_{ii}^2 + 2\sum_{i<j} |h_{ij}|^2\right) = -\frac{1}{2}\operatorname{Tr}(H^2).$$

Thus $f(H) = C_N^{-1} e^{-\frac{1}{2}\operatorname{Tr}(H^2)}$, where $C_N = (2\pi)^{N/2}\pi^{N(N-1)/2}$. □

2.4 Unitary Invariance

Definition 2.5. *The unitary group is* $\mathrm{U}(N) = \{U \in M_N(\mathbb{C}) : U^*U = I\}$.

Proposition 2.6. *The* $\mathrm{GUE}(N)$ *distribution is invariant under conjugation by any fixed unitary matrix* $U \in \mathrm{U}(N)$*:*

$$H \mapsto UHU^*.$$

Proof. The map is a linear transformation on Herm_N. Its Jacobian is 1 because it is an isometry with respect to the Hilbert-Schmidt inner product $\langle A, B\rangle = \operatorname{Tr}(AB^*)$. Furthermore,

$$\operatorname{Tr}((UHU^*)^2) = \operatorname{Tr}(UH^2U^*) = \operatorname{Tr}(H^2),$$

by the cyclicity of the trace. Since the density depends only on $\operatorname{Tr}(H^2)$, it is invariant. □

2.5 Joint Density of Eigenvalues

The following theorem is fundamental to random matrix theory.

Theorem 2.7. *The joint probability density function of the (real) eigenvalues* $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$ *of a matrix* $H \in \mathrm{GUE}(N)$ *is given by*

$$p_N(\lambda_1, \ldots, \lambda_N) = \frac{1}{Z_N} \prod_{1\le i<j\le N} (\lambda_i - \lambda_j)^2 \exp\left(-\frac{1}{2}\sum_{k=1}^N \lambda_k^2\right),$$

where Z_N is the normalization constant (partition function).

Proof. We perform a change of variables from the matrix entries H to its eigenvalues $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_N)$ and eigenvectors $U \in \mathrm{U}(N)$. Let $H = U\Lambda U^*$. Differentiating this relation yields

$$\mathrm{d}H = \mathrm{d}U\Lambda U^* + U\,\mathrm{d}\Lambda U^* + U\Lambda\,\mathrm{d}U^*.$$

Multiplying on the left by U^* and on the right by U, we obtain

$$U^* \, \mathrm{d}HU = U^* \, \mathrm{d}U\Lambda + \, \mathrm{d}\Lambda + \Lambda \, \mathrm{d}U^*U.$$

Define the anti-Hermitian matrix $\Omega = U^* \, \mathrm{d}U$. Differentiating $U^*U = I$ gives $\mathrm{d}U^*U + U^* \, \mathrm{d}U = 0$, so $\mathrm{d}U^*U = -\Omega$ and $\Omega^* = -\Omega$. Substituting,

$$U^* \, \mathrm{d}HU = \, \mathrm{d}\Lambda + [\Omega, \Lambda].$$

In components, this means

$$(U^* \, \mathrm{d}HU)_{ii} = \, \mathrm{d}\lambda_i, \quad (U^* \, \mathrm{d}HU)_{ij} = (\lambda_j - \lambda_i)\Omega_{ij} \quad (i \neq j).$$

The map $H \mapsto (\Lambda, U)$ is a change of variables. The Jacobian factor comes from the transformation of the volume element. Since the map is a submersion with fibers isomorphic to the unitary group, we have

$$\mathrm{d}H = J(\lambda) \prod_{i=1}^{N} \mathrm{d}\lambda_i \, \mathrm{d}\mu(U),$$

where $\mathrm{d}\mu(U)$ is the Haar measure on $\mathrm{U}(N)$. To compute $J(\lambda)$, note that the off-diagonal components satisfy $|\lambda_j - \lambda_i|^2$ for each pair (i, j). A standard computation (see [**mehta2004random**]) shows that

$$J(\lambda) = C'_N \prod_{i<j} (\lambda_i - \lambda_j)^2 = C'_N |\Delta(\lambda)|^2,$$

where $\Delta(\lambda) = \prod_{i<j}(\lambda_j - \lambda_i)$ is the Vandermonde determinant. Substituting this into the Gaussian weight,

$$e^{-\frac{1}{2}\operatorname{Tr}(H^2)} = e^{-\frac{1}{2}\operatorname{Tr}(U\Lambda^2U^*)} = e^{-\frac{1}{2}\sum \lambda_i^2}.$$

Integrating over the unitary group gives a constant factor. Thus,

$$p_N(\lambda_1, \ldots, \lambda_N) = \frac{1}{Z_N} |\Delta(\lambda)|^2 e^{-\frac{1}{2}\sum \lambda_i^2},$$

where Z_N is the normalization constant. The Mehta integral gives

$$Z_N = \int_{\mathbb{R}^N} |\Delta(\lambda)|^2 e^{-\frac{1}{2}\sum \lambda_i^2} \prod_{i=1}^{N} \mathrm{d}\lambda_i = (2\pi)^{N/2} \prod_{j=1}^{N} j!.$$

□

2.6 Coulomb Gas Interpretation

The density can be written as $p_N(\lambda) \propto e^{-\mathcal{E}(\lambda)}$, where

$$\mathcal{E}(\lambda) = \frac{1}{2} \sum_{i=1}^{N} \lambda_i^2 - 2 \sum_{i<j} \log |\lambda_i - \lambda_j|.$$

This is the Boltzmann factor for a one-dimensional system of N charged particles (with logarithmic repulsion) confined by a harmonic potential. The factor 2 in front of the sum over logs indicates the strength of the repulsion.

2.7 Correlation Functions and the GUE Kernel

The n-point correlation functions are defined as marginal densities:

$$R_n^{(N)}(x_1, \ldots, x_n) = \frac{N!}{(N-n)!} \int_{\mathbb{R}^{N-n}} p_N(\lambda_1, \ldots, \lambda_N) \, \mathrm{d}\lambda_{n+1} \ldots \mathrm{d}\lambda_N.$$

For $n = 1$, $R_1^{(N)}(x)$ is the eigenvalue density. For $n = 2$, $R_2^{(N)}(x, y)$ gives the pair correlation density.

The Gaudin-Mehta theorem expresses these correlation functions as determinants of a kernel.

Theorem 2.8 (Gaudin-Mehta)**.** *The n-point correlation functions for* GUE(N) *are given by*

$$R_n^{(N)}(x_1, \ldots, x_n) = \det_{1 \le i,j \le n} \left(K_N(x_i, x_j) \right),$$

where $K_N(x, y)$ is the GUE kernel,

$$K_N(x, y) = \sum_{k=0}^{N-1} \psi_k(x) \psi_k(y),$$

and ψ_k are the normalized Hermite functions:

$$\psi_k(x) = (-1)^k (2^k k! \sqrt{\pi})^{-1/2} e^{x^2/2} \frac{d^k}{dx^k} e^{-x^2}.$$

Proof. Let M be the $N \times N$ matrix with entries $M_{jk} = \psi_{j-1}(x_k)$. Then using the Vandermonde determinant and the properties of Hermite polynomials, one shows that

$$p_N(x_1, \ldots, x_N) = \frac{1}{N!} \det(M^T M).$$

Expanding the determinant,

$$\det(M^T M) = \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma) \prod_{j=1}^{N} \sum_{k=1}^{N} \psi_{j-1}(x_k) \psi_{\sigma(j)-1}(x_k).$$

Using the orthonormality of the Hermite functions,

$$\int_{\mathbb{R}} K_N(x, y) K_N(y, z) \, \mathrm{d}y = K_N(x, z), \quad \int_{\mathbb{R}} K_N(x, x) \, \mathrm{d}x = N.$$

Applying the Gaudin lemma (Lemma A.3) N times yields

$$\int_{\mathbb{R}^N} \det(M^T M) \, \mathrm{d}x_1 \ldots \mathrm{d}x_N = N!.$$

Thus $p_N(\vec{x}) = \frac{1}{N!} \det_{N \times N}(K_N(x_j, x_k))$. The formula for $R_n^{(N)}$ follows by integrating out $N - n$ variables and applying the Gaudin lemma repeatedly. $\square$

2.8 Dyson's Theorem and the Sine Kernel

In the large-N limit, with a suitable rescaling of the eigenvalues in the bulk of the spectrum, the local statistics become universal.

Theorem 2.9 (Dyson's Theorem)**.** *For $n \geq 2$, the limiting n-point correlation function in the bulk is*

$$\lim_{N\to\infty} \left(\frac{\pi}{\sqrt{2N}}\right)^n R_n^{(N)} \left(\frac{\pi x_1}{\sqrt{2N}}, \ldots, \frac{\pi x_n}{\sqrt{2N}}\right) = \det_{1\leq i,j\leq n} \left(S(x_i - x_j)\right),$$

where $S(x)$ is the Dyson sine kernel:

$$S(x) = \frac{\sin(\pi x)}{\pi x}.$$

Proof. Using the Christoffel-Darboux formula (Proposition B.5), we have for $x \neq y$,

$$K_N(x,y) = \sqrt{\frac{N}{2}} \frac{\psi_N(x)\psi_{N-1}(y) - \psi_{N-1}(x)\psi_N(y)}{x-y}.$$

Set $x = \frac{\pi u}{\sqrt{2N}}$, $y = \frac{\pi v}{\sqrt{2N}}$. The asymptotic formula for Hermite functions (see Szegő [**szego1939orthogonal**]) gives as $N \to \infty$,

$$\psi_{2m}\left(\frac{\pi u}{\sqrt{2N}}\right) \sim \frac{(-1)^m}{\sqrt{\pi} m^{1/4}} \cos(\pi u), \quad \psi_{2m+1}\left(\frac{\pi u}{\sqrt{2N}}\right) \sim \frac{(-1)^m}{\sqrt{\pi} m^{1/4}} \sin(\pi u).$$

Substituting and simplifying,

$$\frac{\pi}{\sqrt{2N}} K_N\left(\frac{\pi u}{\sqrt{2N}}, \frac{\pi v}{\sqrt{2N}}\right) \to \frac{\sin(\pi(u-v))}{\pi(u-v)} = S(u-v).$$

The result follows from the determinantal structure of the correlation functions. □

For $n = 2$, this gives the explicit pair correlation function:

$$r_2^{\mathrm{GUE}}(x_1, x_2) = 1 - \left(\frac{\sin(\pi(x_1 - x_2))}{\pi(x_1 - x_2)}\right)^2.$$

2.9 Gap Distributions

Let $E_k^{\mathrm{GUE}(N)}(\sigma)$ be the probability that exactly k eigenvalues lie in an interval of length σ, and $\mu_k^{\mathrm{GUE}(N)}(\sigma)$ the probability density for k-consecutive eigenvalue gaps.

Proposition 2.10. *For $k \geq 1$,*

$$E_k^{\mathrm{GUE}(N)}(\sigma) = \frac{1}{k!} \frac{\partial^k}{\partial T^k} \det\left(I + T\chi_{(0,\sigma)} K_{\mathrm{GUE}(N)}\right)\Big|_{T=-1},$$

$$\mu_k^{\mathrm{GUE}(N)}(\sigma) = \frac{1}{\rho_1^{(N)}(0)} \frac{\partial^2}{\partial \sigma^2} \sum_{j=0}^{k} (k-j) E_j^{\mathrm{GUE}(N)}(\sigma).$$

Proof. The probability that exactly k eigenvalues lie in $[0, \sigma]$ is given by an inclusion-exclusion sum over m-point correlation functions:

$$E_k^{\mathrm{GUE}(N)}(\sigma) = \sum_{m=0}^{N-k} \frac{(-1)^m}{m!} \int_{[0,\sigma]^m} R_{k+m}^{(N)}(x_1, \ldots, x_{k+m}) \prod_{i=1}^{m} \mathrm{d}x_i.$$

Using the determinantal form $R_{k+m}^{(N)}(x_1, \ldots, x_{k+m}) = \det(K_N(x_i, x_j))_{i,j=1}^{k+m}$, this sum is exactly the Taylor expansion of the Fredholm determinant:

$$\det(I + T\chi_{[0,\sigma]} K_N) = \sum_{n=0}^{\infty} \frac{T^n}{n!} \int_{[0,\sigma]^n} \det(K_N(x_i, x_j))_{i,j=1}^{n} \prod_{i=1}^{n} \mathrm{d}x_i.$$

The coefficient of T^{k+m} gives the required term. The formula for μ_k follows from the relation between the gap probability and the distribution of the $(k+1)$-st eigenvalue. □

In the large-N limit, with $\sigma = \pi\eta/\sqrt{2N}$,

$$\lim_{N\to\infty} E_k^{\mathrm{GUE}(N)}\left(\frac{\pi\eta}{\sqrt{2N}}\right) = E_k^{\mathrm{GUE}}(\eta) = \frac{1}{k!} \frac{\partial^k}{\partial T^k} \det(I + T\chi_{[0,\eta]} S)\Big|_{T=-1},$$

$$\lim_{N\to\infty} \frac{\pi}{\sqrt{2N}} \mu_k^{\mathrm{GUE}(N)}\left(\frac{\pi\eta}{\sqrt{2N}}\right) = \mu_k^{\mathrm{GUE}}(\eta) = \frac{\partial^2}{\partial \eta^2} \sum_{j=0}^{k} (k-j) E_j^{\mathrm{GUE}}(\eta).$$

Numerically, the Gaudin distribution $\mu_1^{\mathrm{GUE}}(\eta)$ has the following values:

$$\mu_1^{\mathrm{GUE}}(0.5) \approx 0.726, \quad \mu_1^{\mathrm{GUE}}(1.0) \approx 0.509, \quad \mu_1^{\mathrm{GUE}}(1.5) \approx 0.302, \quad \mu_1^{\mathrm{GUE}}(2.0) \approx 0.144.$$

3 Random Matrices: Classical Compact Groups

3.1 Definitions and Haar Measure

Definition 3.1. *The classical compact groups are:*

- *The unitary group* $\mathrm{U}(N) = \{X \in M_N(\mathbb{C}) : X^* X = I\}$.
- *The special orthogonal group* $\mathrm{SO}(N) = \{X \in M_N(\mathbb{R}) : X^T X = I, \det X = 1\}$.
- *The symplectic group* $\mathrm{USp}(2N) = \{X \in \mathrm{U}(2N) : X^T J X = J\}$, *where* $J = \begin{pmatrix} 0 & -I_N \\ I_N & 0 \end{pmatrix}$.

Each carries a unique normalized Haar measure $dP_{G(N)}$.

The Weyl integration formula expresses integrals of class functions over these groups as integrals over their eigenvalues.

Theorem 3.2 (Weyl Integration Formula)**.** *For a class function* f *on* $G(N)$,

$$\int_{G(N)} f(X)\, dP_{G(N)}(X) = \int_{[0,\sigma\pi]^N} f(\theta_1, \ldots, \theta_N) e_{G(N)}(\theta_1, \ldots, \theta_N)\, d\theta_1 \cdots d\theta_N,$$

with the densities:

$$e_{\mathrm{U}(N)}(\theta_1,\dots,\theta_N) = \frac{1}{(2\pi)^N N!}\prod_{1\le j<k\le N}|e^{i\theta_k}-e^{i\theta_j}|^2,$$

$$e_{\mathrm{SO}(2N)}(\theta_1,\dots,\theta_N) = \frac{2^{(N-1)^2}}{\pi^N N!}\prod_{1\le j<k\le N}(\cos\theta_k-\cos\theta_j)^2,$$

$$e_{\mathrm{SO}(2N+1)}(\theta_1,\dots,\theta_N) = \frac{2^{N^2}}{\pi^N N!}\prod_{1\le j<k\le N}(\cos\theta_k-\cos\theta_j)^2\prod_{h=1}^{N}\sin^2\left(\frac{\theta_h}{2}\right),$$

$$e_{\mathrm{USp}(2N)}(\theta_1,\dots,\theta_N) = \frac{2^{N^2}}{\pi^N N!}\prod_{1\le j<k\le N}(\cos\theta_k-\cos\theta_j)^2\prod_{h=1}^{N}\sin^2(\theta_h).$$

Here $\sigma = 2$ *for* $\mathrm{U}(N)$ *and* $\sigma = 1$ *otherwise.*

Proof for $\mathrm{U}(N)$. We give a complete proof for $\mathrm{U}(N)$; the other cases are analogous but with different parameterizations of the eigenvalues. Let $dP_{\mathrm{U}(N)}(X)$ be the normalized Haar measure. For a class function f, we have

$$\int_{\mathrm{U}(N)} f(X)\,dP_{\mathrm{U}(N)}(X) = \int_T f(t)\,d\mu_T(t),$$

where $T \cong \mathrm{U}(1)^N$ is a maximal torus consisting of diagonal matrices $t = \mathrm{diag}(e^{i\theta_1},\dots,e^{i\theta_N})$, and $d\mu_T$ is the pushforward of Haar measure to T. The Weyl denominator formula gives

$$d\mu_T(\theta_1,\dots,\theta_N) = \frac{1}{N!}\prod_{1\le j<k\le N}|e^{i\theta_k}-e^{i\theta_j}|^2\frac{d\theta_1\cdots d\theta_N}{(2\pi)^N}.$$

This factor is the Jacobian of the conjugation map $G\times T\to G$. Integrating over the Weyl group accounts for the $N!$ factor. Thus

$$\int_{\mathrm{U}(N)} f(X)\,dP_{\mathrm{U}(N)}(X) = \frac{1}{(2\pi)^N N!}\int_{[0,2\pi]^N} f(\theta_1,\dots,\theta_N)\prod_{j<k}|e^{i\theta_k}-e^{i\theta_j}|^2\prod_{j=1}^{N}d\theta_j.$$

This is exactly the stated formula. □

3.2 Determinantal Structure

For each group, the eigenvalue density can be written as a determinant involving a kernel function $K_{G(N)}(\theta,\phi)$. For $\mathrm{U}(N)$,

$$K_{\mathrm{U}(N)}(\theta,\phi) = \frac{1}{2\pi}\frac{\sin\left(\frac{N}{2}(\theta-\phi)\right)}{\sin\left(\frac{1}{2}(\theta-\phi)\right)}.$$

For $\mathrm{SO}(2N)$,

$$K_{\mathrm{SO}(2N)}(\theta,\phi) = \frac{1}{2\pi}\left(\frac{\sin\left((N-\frac{1}{2})(\theta-\phi)\right)}{\sin\left(\frac{1}{2}(\theta-\phi)\right)}+\frac{\sin\left((N-\frac{1}{2})(\theta+\phi)\right)}{\sin\left(\frac{1}{2}(\theta+\phi)\right)}\right).$$

For $\mathrm{USp}(2N)$,

$$K_{\mathrm{USp}(2N)}(\theta,\phi) = \frac{1}{2\pi}\left(\frac{\sin\left(N(\theta-\phi)\right)}{\sin\left(\frac{1}{2}(\theta-\phi)\right)}-\frac{\sin\left(N(\theta+\phi)\right)}{\sin\left(\frac{1}{2}(\theta+\phi)\right)}\right).$$

Theorem 3.3. *For any of these groups,*

$$e_{G(N)}(\theta_1, \ldots, \theta_N) = \frac{1}{(\sigma\pi)^N N!} \det_{N\times N} \left(K_{G(N)}(\theta_j, \theta_k) \right).$$

Proof. The proof uses the transpose lemma (Lemma A.2) and trigonometric identities to express the products in the Weyl density as determinants. For $\mathrm{U}(N)$, note that

$$\prod_{j<k} |e^{i\theta_k} - e^{i\theta_j}|^2 = \det(e^{i(j-1)\theta_k})_{j,k} \det(e^{-i(j-1)\theta_k})_{j,k} = \det\left(\sum_{l=1}^{N} e^{i(l-1)(\theta_k - \theta_j)} \right).$$

The sum is a geometric series, which equals $e^{i(N-1)(\theta_k-\theta_j)/2} S_N(\theta_k - \theta_j)$. Absorbing the phase factors into rows and columns yields the determinant of $S_N(\theta_k - \theta_j)$. A similar computation using Chebyshev polynomials gives the kernels for the orthogonal and symplectic cases. □

3.3 n-Level Correlation and Density

Using the Gaudin-Mehta theorem, the n-level correlation functions are

$$R_n^{G(N)}(x_1, \ldots, x_n) = \det_{1\le i,j\le n} \left(K_{G(N)}(x_i, x_j) \right).$$

In the large-N limit, after appropriate rescaling, the n-level correlations for $\mathrm{U}(N)$ converge to the GUE limit.

Theorem 3.4 (Limiting 1-Level Densities)**.** *As $N \to \infty$, the rescaled 1-level densities converge to:*

$$W_{1,\mathrm{U}}(x) = 1, \tag{1}$$

$$W_{1,\mathrm{USp}}(x) = 1 - \frac{\sin 2\pi x}{2\pi x}, \tag{2}$$

$$W_{1,\mathrm{SO(even)}}(x) = 1 + \frac{\sin 2\pi x}{2\pi x}, \tag{3}$$

$$W_{1,\mathrm{SO(odd)}}(x) = 1 - \frac{\sin 2\pi x}{2\pi x} + \delta_0(x), \tag{4}$$

$$W_{1,\mathrm{O}}(x) = 1 + \frac{1}{2}\delta_0(x). \tag{5}$$

Proof for $\mathrm{U}(N)$*.* From Theorem 2.22 with $n = 1$, we have

$$\lim_{N\to\infty} \int_{\mathrm{U}(N)} \sum_{j=1}^{N} f(\bar{\theta}_j)\, dP_{\mathrm{U}(N)}(X) = \int_0^\infty f(x)\, dx,$$

since $K_U(x, x) = 1$ and the determinant is trivial. This shows $W_{1,\mathrm{U}}(x) = 1$. For the other groups, one computes the limit of the kernel $K_{G(N)}$ explicitly using the asymptotic of $S_{\rho N+\tau}(\theta)$ as $N \to \infty$ with the scaling $\theta = \frac{\sigma\pi x}{N}$. For $\mathrm{USp}(2N)$, one obtains $K_{\mathrm{USp}}(x, y) = S(y - x) - S(y + x)$, whose diagonal gives $W_{1,\mathrm{USp}}(x)$. For $\mathrm{SO}(2N + 1)$, the δ_0 term arises from the eigenvalue at 1. □

3.4 Gap Distributions

For the classical compact groups, the gap distributions coincide with the Gaudin distributions from GUE. This is a consequence of the universality of local bulk statistics.

Theorem 3.5. *For each $k \geq 1$, as $N \to \infty$,*

$$\int_{G(N)} \mu_k^{G(N)}(s)\, dP_{G(N)}(X) \to \mu_k^{\mathrm{GUE}}(s),$$

where μ_k^{GUE} are the Gaudin measures.

Proof. The proof follows from expressing $\mu_k^{G(N)}(s)$ in terms of n-level correlations via inclusion-exclusion (Lemma 2.18 and Corollary 2.19), then using the limiting behavior of the n-level correlations (Theorem 2.15) and the dominated convergence theorem. □

3.5 Distribution of the First Eigenvalue

For low-lying eigenvalues (near 1), the distributions are symmetry-dependent.

Theorem 3.6. *For $j \geq 1$, there exist probability measures ν_j^G on $[0, \infty)$ such that*

$$\lim_{N\to\infty} \nu_j^{G(N)}(s) = \nu_j^G(s).$$

For $j = 1$, the densities expand as:

$$\nu_1^{\mathrm{U}}(s) = 1 - \frac{\pi^2}{9}s^3 + O(s^5),$$
$$\nu_1^{\mathrm{SO(even)}}(s) = 2 - \frac{2\pi^2}{3}s^2 + O(s^4),$$
$$\nu_1^{\mathrm{USp}}(s) = \frac{\pi^2}{3}s^2 + O(s^4).$$

Proof. Let $E_n^{G(N)}(s)$ be the probability of exactly n eigenvalues in $[0, s]$. Then

$$E_n^{G(N)}(s) = \frac{1}{n!}\frac{\partial^n}{\partial T^n} \det(I + T\chi_{[0,s]}K_{G(N)})\big|_{T=-1}.$$

The relation $\{\theta_j > s\} = \bigcup_{n=0}^{j-1}\{\text{exactly } n \text{ eigenvalues in } [0, s]\}$ gives

$$\int_s^\infty \nu_j^{G(N)}(x)\,\mathrm{d}x = \sum_{n=0}^{j-1} E_n^{G(N)}(s),$$

so $\nu_j^{G(N)}(s) = -\frac{d}{ds}\sum_{n=0}^{j-1} E_n^{G(N)}(s)$. Taking the limit $N \to \infty$ and using the convergence of the Fredholm determinants (Theorem B.8) yields the result. The asymptotic expansions are obtained by expanding the Fredholm determinant for small s. □

4 Pair Correlation of Zeros of the Riemann Zeta Function

This chapter provides a complete and detailed proof of Montgomery's theorem on the pair correlation of the zeros of the Riemann zeta function. The proof follows the original approach of Montgomery [**montgomery1973pair**], which is based on the "dual" version of the explicit formula. We assume the Riemann Hypothesis (RH) throughout, which states that all non-trivial zeros are of the form $\rho = 1/2 + \mathrm{i}\gamma$ with $\gamma \in \mathbb{R}$.

4.1 The Pair Correlation Function and a Key Proposition

We begin by defining the central object of study, the pair correlation function.

Definition 4.1. *Let $T \geq 2$. Define the function $F(\alpha, T)$ for a real parameter α by*

$$F(\alpha, T) := \left(\frac{T}{2\pi}\log T\right)^{-1} \sum_{0<\gamma,\gamma'\leq T} T^{\mathrm{i}\alpha(\gamma-\gamma')} \frac{4}{4+(\gamma-\gamma')^2}, \tag{6}$$

where the sum runs over the ordinates γ of the non-trivial zeros of $\zeta(s)$.

The factor $4/(4+(\gamma-\gamma')^2)$ acts as a smooth weight, ensuring convergence and making the function analytic in α. The scaling factor $\frac{T}{2\pi}\log T$ is approximately the number of zeros $N(T) \sim \frac{T}{2\pi}\log T$, so $F(0, T)$ is essentially the average of the weights over all pairs.

The following proposition shows that $F(\alpha, T)$ is the Fourier transform of the weighted pair correlation.

Proposition 4.2. *Let f be a Schwartz function on $\mathbb{R}$. Then*

$$\sum_{0<\gamma,\gamma'\leq T} f\left(\frac{\log T}{2\pi}(\gamma-\gamma')\right) \frac{4}{4+(\gamma-\gamma')^2} = \frac{T}{2\pi}\log T \int_{-\infty}^{\infty} F(\alpha, T)\hat{f}(\alpha)\ \mathrm{d}\alpha.$$

Proof. Starting from the definition of $F(\alpha, T)$,

$$\frac{T}{2\pi}\log T \int_{-\infty}^{\infty} F(\alpha, T)\hat{f}(\alpha)\ \mathrm{d}\alpha = \int_{-\infty}^{\infty} \sum_{\gamma,\gamma'} T^{\mathrm{i}\alpha(\gamma-\gamma')} \frac{4}{4+(\gamma-\gamma')^2}\hat{f}(\alpha)\ \mathrm{d}\alpha.$$

Interchanging sum and integral (justified by absolute convergence),

$$= \sum_{\gamma,\gamma'} \frac{4}{4+(\gamma-\gamma')^2} \int_{-\infty}^{\infty} \hat{f}(\alpha) e^{2\pi\mathrm{i}\alpha\frac{\log T}{2\pi}(\gamma-\gamma')}\ \mathrm{d}\alpha.$$

By Fourier inversion, $\int_{-\infty}^{\infty} \hat{f}(\alpha)e^{2\pi\mathrm{i}\alpha x}\ \mathrm{d}\alpha = f(x)$. Thus,

$$= \sum_{\gamma,\gamma'} f\left(\frac{\log T}{2\pi}(\gamma-\gamma')\right) \frac{4}{4+(\gamma-\gamma')^2}.$$

□

Thus, understanding the limiting behavior of the pair correlation is equivalent to understanding the limit of $F(\alpha, T)$ as $T \to \infty$.

4.2 The Explicit Formula for the Riemann Zeta Function

We require a version of the explicit formula that isolates the contribution of zeros with a specific weight.

Proposition 4.3 (A Weighted Explicit Formula)**.** *Assume RH. Let* $1 < \sigma < 2$*,* $x \geq 1$*, and* $t \in \mathbb{R}$*. Then*

$$\begin{aligned}(2\sigma-1)\sum_{\gamma}\frac{x^{\mathrm{i}\gamma}}{(\sigma-\frac{1}{2})^2+(t-\gamma)^2} &= -x^{-\frac{1}{2}}\left[\sum_{n\leq x}\Lambda(n)\left(\frac{x}{n}\right)^{1-\sigma+\mathrm{i}t}+\sum_{n>x}\Lambda(n)\left(\frac{x}{n}\right)^{\sigma+\mathrm{i}t}\right]\\ &\quad - x^{\frac{1}{2}-\sigma+\mathrm{i}t}\frac{\zeta'}{\zeta}(1-\sigma+\mathrm{i}t)\\ &\quad + \frac{x^{\frac{1}{2}}(2\sigma-1)}{(\sigma-1+\mathrm{i}t)(\sigma-\mathrm{i}t)}\\ &\quad - x^{-\frac{1}{2}}\sum_{n=1}^{\infty}\frac{x^{-2n}(2\sigma-1)}{(2n+\sigma+\mathrm{i}t)(\sigma-1-2n-\mathrm{i}t)}. \end{aligned} \tag{7}$$

Proof. Start from Riemann's explicit formula for $\mathrm{Re}(s) > 1$:

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}.$$

The completed formula, valid for $x > 1$, $x \neq p^n$, is

$$\sum_{n\leq x}\frac{\Lambda(n)}{n^s} = -\frac{\zeta'}{\zeta}(s) + \frac{x^{1-s}}{1-s} - \sum_{\rho}\frac{x^{\rho-s}}{\rho-s} + \sum_{n=1}^{\infty}\frac{x^{-2n-s}}{2n+s}.$$

Apply this for $s = \sigma + \mathrm{i}t$ and for $s = 1 - \sigma + \mathrm{i}t$. Under RH, $\rho = 1/2 + \mathrm{i}\gamma$. Write the sum over zeros for both values:

$$\begin{aligned} S_1 &:= \sum_{\gamma}\frac{x^{1/2+\mathrm{i}\gamma-s}}{1/2+\mathrm{i}\gamma-s} = -\frac{\zeta'}{\zeta}(s) - \sum_{n\leq x}\frac{\Lambda(n)}{n^s} + \frac{x^{1-s}}{1-s} + \sum_{n=1}^{\infty}\frac{x^{-2n-s}}{2n+s},\\ S_2 &:= \sum_{\gamma}\frac{x^{1/2+\mathrm{i}\gamma-(1-s)}}{1/2+\mathrm{i}\gamma-(1-s)} = -\frac{\zeta'}{\zeta}(1-s) - \sum_{n\leq x}\frac{\Lambda(n)}{n^{1-s}} + \frac{x^s}{s} + \sum_{n=1}^{\infty}\frac{x^{-2n-1+s}}{2n+1-s}. \end{aligned}$$

Now note the algebraic identity:

$$\frac{1}{1/2+\mathrm{i}\gamma-s} - \frac{1}{1/2+\mathrm{i}\gamma-(1-s)} = \frac{2\sigma-1}{(\sigma-1/2)^2+(t-\gamma)^2}.$$

Multiplying by $x^{1/2+\mathrm{i}\gamma}$ and summing over γ,

$$(2\sigma-1)\sum_{\gamma}\frac{x^{\mathrm{i}\gamma}}{(\sigma-1/2)^2+(t-\gamma)^2} = x^{\sigma-1/2}(S_1 - S_2).$$

Substitute S_1 and S_2, multiply by $x^{1/2-\sigma}$, and simplify to obtain the stated formula. □

For our purposes, we set $\sigma = 3/2$. This simplifies $2\sigma - 1 = 2$ and the exponents involving x.

Corollary 4.4. *Assume RH. For* $x \geq 1$ *and* $t \in \mathbb{R}$*,*

$$\begin{aligned} 2\sum_{\gamma}\frac{x^{\mathrm{i}\gamma}}{1+(t-\gamma)^2} &= -x^{-\frac{1}{2}}\left[\sum_{n\leq x}\Lambda(n)\left(\frac{x}{n}\right)^{-\frac{1}{2}+\mathrm{i}t}+\sum_{n>x}\Lambda(n)\left(\frac{x}{n}\right)^{\frac{3}{2}+\mathrm{i}t}\right]\\ &\quad + x^{-1+\mathrm{i}t}\left(\log(|t|+2)+O(1)\right) + O\left(\frac{x^{\frac{1}{2}}}{|t|+2}\right). \end{aligned} \tag{8}$$

Proof. For $\sigma = 3/2$, we have $1-\sigma = -1/2$. Using the bound $\frac{\zeta'}{\zeta}(1-\sigma+\mathrm{i}t) = -\log(|t|+2) + O(1)$ for $|t|$ large (see Davenport [**davenport2000multiplicative**]), and estimating the remaining terms gives the result. □

Denote the left side by $L(x,t)$ and the right side by $R(x,t)$.

4.3 Analysis of $\int_0^T |L(x,t)|^2 \,\mathrm{d}t$

We now compute the mean square of $L(x,t)$.

Lemma 4.5. *For $T \geq 2$ and $t \in [0,T]$,*

$$\sum_{\gamma\in[0,T]} \frac{1}{1+(t-\gamma)^2} \ll \left(\frac{1}{t+1} + \frac{1}{T-t+1}\right)\log T,$$

and

$$\sum_{\gamma>0} \frac{1}{1+(t-\gamma)^2} \ll \log T.$$

Proof. Using $N(T+1) - N(T) = O(\log T)$ (see Lemma 3.13),

$$\sum_{\gamma\in[0,T]} \frac{1}{1+(t-\gamma)^2} \ll \int_0^T \frac{1}{1+(t-u)^2} dN(u) \ll \log T \int_0^T \frac{du}{1+(t-u)^2}.$$

Evaluating the integral gives $\int_0^T \frac{du}{1+(t-u)^2} = \arctan(t) + \arctan(T-t) \ll \frac{1}{t+1} + \frac{1}{T-t+1}$. The second bound follows by splitting the sum at T. □

Proposition 4.6. *For $x = T^\alpha$ with $\alpha \in \mathbb{R}$,*

$$\int_0^T |L(x,t)|^2 \,\mathrm{d}t = 4 \sum_{\gamma,\gamma'\in[0,T]} x^{\mathrm{i}(\gamma-\gamma')} \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{(1+(t-\gamma)^2)(1+(t-\gamma')^2)} + O(\log^3 T).$$

Proof. Write $L(x,t) = 2\sum_\gamma \frac{x^{\mathrm{i}\gamma}}{1+(t-\gamma)^2}$. Then

$$|L(x,t)|^2 = 4\sum_{\gamma,\gamma'} \frac{x^{\mathrm{i}(\gamma-\gamma')}}{(1+(t-\gamma)^2)(1+(t-\gamma')^2)}.$$

Split the double sum into $\gamma,\gamma' \in [0,T]$ and the rest. Using Lemma 4.5, the contribution from pairs with at least one zero outside $[0,T]$ is bounded by

$$\ll \int_0^T \left(\sum_{\gamma\in[0,T]} \frac{1}{1+(t-\gamma)^2}\right)\left(\sum_{\gamma'\notin[0,T]} \frac{1}{1+(t-\gamma')^2}\right) \mathrm{d}t \ll \log T \cdot \log T \cdot \int_0^T \left(\frac{1}{t+1} + \frac{1}{T-t+1}\right) \mathrm{d}t.$$

The integral is $O(\log T)$, so the contribution is $O(\log^3 T)$. Extending the integral to $\mathbb{R}$ introduces an error $O(\log^2 T)$ from t outside $[0,T]$, giving a total error $O(\log^3 T)$. □

Lemma 4.7. *For $\gamma,\gamma' \in \mathbb{R}$,*

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}t}{(1+(t-\gamma)^2)(1+(t-\gamma')^2)} = \frac{\pi}{2}\frac{4}{4+(\gamma-\gamma')^2}.$$

Proof. Consider $f(z) = \frac{1}{(1+(z-\gamma)^2)(1+(z-\gamma')^2)}$. The poles are at $z = \gamma \pm \mathrm{i}$ and $z = \gamma' \pm \mathrm{i}$. For the contour $\Gamma_R = [-R, R] \cup \{Re^{i\theta} : 0 \le \theta \le \pi\}$, the integral over the semicircle vanishes as $R \to \infty$. The poles in the upper half-plane are $z_1 = \gamma + \mathrm{i}$ and $z_2 = \gamma' + \mathrm{i}$. Their residues are

$$\mathrm{Res}(f, \gamma + \mathrm{i}) = \frac{-1}{2(\gamma - \gamma')(2 - \mathrm{i}(\gamma - \gamma'))}, \quad \mathrm{Res}(f, \gamma' + \mathrm{i}) = \frac{-1}{2(\gamma' - \gamma)(2 - \mathrm{i}(\gamma' - \gamma))}.$$

Summing and multiplying by $2\pi\mathrm{i}$ gives

$$\int_{-\infty}^{\infty} f(t)\,\mathrm{d}t = \frac{\pi}{2}\frac{4}{4 + (\gamma - \gamma')^2}.$$

□

Combining Proposition 4.6 with Lemma 4.7 and setting $x = T^\alpha$ yields

$$\int_0^T |L(T^\alpha, t)|^2\,\mathrm{d}t = 2\pi \sum_{\gamma,\gamma' \in [0,T]} T^{\mathrm{i}\alpha(\gamma-\gamma')} \frac{4}{4 + (\gamma - \gamma')^2} + O(\log^3 T) = F(\alpha, T) \cdot T \log T + O(\log^3 T).$$

4.4 Analysis of $\int_0^T |R(x,t)|^2\,\mathrm{d}t$

Recall $R(x,t) = A(x,t) + B(x,t) + C(x,t) + D(x,t)$ where

$$\begin{aligned}
A(x,t) &= -x^{-\frac{1}{2}} \left[\sum_{n \le x} \Lambda(n) \left(\frac{x}{n}\right)^{-\frac{1}{2}+\mathrm{i}t} + \sum_{n > x} \Lambda(n) \left(\frac{x}{n}\right)^{\frac{3}{2}+\mathrm{i}t} \right], \\
B(x,t) &= x^{-1+\mathrm{i}t} \log(|t| + 2), \\
C(x,t) &= O(x^{-1+\mathrm{i}t}), \\
D(x,t) &= O\left(\frac{x^{\frac{1}{2}}}{|t| + 2}\right).
\end{aligned}$$

Lemma 4.8. *For $T \ge 2$ and $x \ge 1$,*

$$\int_0^T |B(x,t)|^2\,\mathrm{d}t \ll \frac{T}{x^2}\log^2 T, \quad \int_0^T |C(x,t)|^2\,\mathrm{d}t \ll \frac{T}{x^2}, \quad \int_0^T |D(x,t)|^2\,\mathrm{d}t \ll x.$$

Proof. $|B(x,t)|^2 = x^{-2}\log^2(|t|+2)$. Integrating gives $\frac{T}{x^2}\log^2 T + O(\frac{T}{x^2}\log T)$. Similarly, $|C(x,t)|^2 \ll x^{-2}$, so its integral is $O(T/x^2)$. For D, $|D(x,t)|^2 \ll x/(t+2)^2$, and $\int_0^T \frac{\mathrm{d}t}{(t+2)^2} \ll 1$, so the integral is $O(x)$. □

The main contribution comes from $A(x,t)$. Write $A(x,t) = -x^{-\frac{1}{2}} \sum_{n=1}^{\infty} \Lambda(n) a_n(x) n^{-\mathrm{i}t}$, where

$$a_n(x) = \begin{cases} \left(\frac{n}{x}\right)^{\frac{1}{2}}, & n \le x, \\ \left(\frac{x}{n}\right)^{\frac{3}{2}}, & n > x. \end{cases}$$

Then $|A(x,t)|^2 = x^{-1} \left|\sum_{n=1}^{\infty} \Lambda(n) a_n(x) n^{-\mathrm{i}t}\right|^2$.

Lemma 4.9 (Quantitative Parseval)**.** *For any sequence b_n with $\sum_n |b_n|^2 n < \infty$,*

$$\int_0^T \left| \sum_{n=1}^{\infty} b_n n^{-\mathrm{i}t} \right|^2 \mathrm{d}t = T \sum_{n=1}^{\infty} |b_n|^2 + O\left(\sum_{n=1}^{\infty} n |b_n|^2\right).$$

Proof. Expand the square:

$$\left|\sum_n b_n n^{-\mathrm{i}t}\right|^2 = \sum_{n,m} b_n \overline{b_m} \left(\frac{n}{m}\right)^{\mathrm{i}t}.$$

Integrate term by term:

$$\int_0^T \left(\frac{n}{m}\right)^{\mathrm{i}t} \mathrm{d}t = \begin{cases} T, & n = m, \\ \frac{(n/m)^{\mathrm{i}T}-1}{\mathrm{i}\log(n/m)}, & n \neq m. \end{cases}$$

The off-diagonal terms contribute $O\left(\sum_{n\neq m} \frac{|b_n b_m|}{|\log(n/m)|}\right)$. Using the bound $|\log(n/m)| \geq c|n-m|/\max(n,m)$, one obtains the estimate. □

Applying this with $b_n = \Lambda(n)a_n(x)$,

$$\int_0^T |A(x,t)|^2 \,\mathrm{d}t = \frac{T}{x}\sum_{n=1}^{\infty} \Lambda(n)^2 a_n(x)^2 + \frac{1}{x}\sum_{n=1}^{\infty} n\Lambda(n)^2 a_n(x)^2.$$

Denote the first term by $A_1(T)$ and the second by $A_2(T)$.

Lemma 4.10. *Assuming RH,*

$$\sum_{n\leq y} \Lambda(n)^2 = y\log y - y + O(y^{\frac{1}{2}}\log^3 y).$$

Proof. Using $\psi(y) = y + O(y^{\frac{1}{2}}\log^2 y)$ (which follows from RH) and summation by parts,

$$\sum_{n\leq y} \Lambda(n)^2 = \int_1^y \log t \, d\psi(t) = \psi(y)\log y - \int_1^y \frac{\psi(t)}{t}\,\mathrm{d}t.$$

Substituting the approximation for ψ gives the result. □

Proposition 4.11. *For $T \geq 2$ and $x \geq 1$,*

$$A_1(T) = T\left(\log x + O(1)\right).$$

Proof. Write $A_1(T) = \frac{T}{x^2}\sum_{n\leq x} n\Lambda(n)^2 + Tx^2\sum_{n>x}\frac{\Lambda(n)^2}{n^3}$. For the first sum, using Lemma 4.10 and summation by parts,

$$\sum_{n\leq x} n\Lambda(n)^2 = \frac{x^2}{2}\log x + O(x^2).$$

Thus $\frac{T}{x^2}\sum_{n\leq x} n\Lambda(n)^2 = \frac{T}{2}\log x + O(T)$. For the second sum,

$$\sum_{n>x}\frac{\Lambda(n)^2}{n^3} = \int_x^{\infty}\frac{1}{t^3}d\left(\sum_{n\leq t}\Lambda(n)^2\right) = \frac{\log x}{2x^2} + O\left(\frac{1}{x^2}\right).$$

Thus $Tx^2\sum_{n>x}\frac{\Lambda(n)^2}{n^3} = \frac{T}{2}\log x + O(T)$. Adding the two contributions gives $T\log x + O(T)$. □

Proposition 4.12. *For $T \geq 2$ and $x \geq 1$,*

$$A_2(T) \ll x\log x.$$

Proof. Write $A_2(T) = \frac{1}{x^2}\sum_{n\le x} n^2\Lambda(n)^2 + x^2\sum_{n>x}\frac{\Lambda(n)^2}{n^2}$. Using Lemma 4.10 and summation by parts,

$$\sum_{n\le x} n^2\Lambda(n)^2 = \frac{x^3}{3}\log x + O(x^3),$$

so the first term is $O(x\log x)$. For the second term,

$$\sum_{n>x}\frac{\Lambda(n)^2}{n^2} = \int_x^\infty \frac{1}{t^2} d\left(\sum_{n\le t}\Lambda(n)^2\right) = \frac{\log x}{x} + O\left(\frac{1}{x}\right),$$

so the second term is $O(x\log x)$. □

Thus,

$$\int_0^T |A(x,t)|^2\,\mathrm{d}t = T\log x + O(T) + O(x\log x).$$

Now we consider three ranges for x relative to T:

(i) $1 \le x \le \log^{3/4} T$.

(ii) $\log^{3/4} T < x \le \log^{3/2} T$.

(iii) $\log^{3/2} T < x \le T/\log T$.

In case (i), the dominant term is B: $M_2 = \frac{T}{x^2}\log^2 T \ge \frac{T}{\log^{3/2} T}\log^2 T = T\log^{1/2} T$, while $M_1 = T\log x + O(T) \ll T\log\log T = o(M_2)$. So

$$\int_0^T |R|^2\,\mathrm{d}t = (1+o(1))\frac{T}{x^2}\log^2 T.$$

In case (ii), all terms are $o(T\log T)$.

In case (iii), $M_1 = T\log x + O(T)$ is dominant, and

$$\int_0^T |R|^2\,\mathrm{d}t = (1+o(1))T\log x.$$

Putting $x = T^\alpha$ with $0 \le \alpha \le 1-\varepsilon$, we obtain

$$\int_0^T |R(T^\alpha,t)|^2\,\mathrm{d}t = \left[(1+o(1))\frac{T}{T^{2\alpha}}\log T + \alpha + o(1)\right] T\log T.$$

4.5 Equating the Two Integrals

From Sections 3.3 and 3.4, we have

$$\int_0^T |L(T^\alpha,t)|^2\,\mathrm{d}t = F(\alpha,T)T\log T + O(\log^3 T),$$

$$\int_0^T |R(T^\alpha,t)|^2\,\mathrm{d}t = \left[(1+o(1))T^{1-2\alpha}\log T + \alpha + o(1)\right] T\log T.$$

Since $L(x,t) = R(x,t)$, the two integrals are equal. Cancelling $T\log T$ and letting $T\to\infty$ gives

$$F(\alpha,T) = (1+o(1))T^{-2\alpha}\log T + \alpha + o(1),$$

uniformly for $0 \le \alpha \le 1-\varepsilon$. By symmetry $F(\alpha,T) = F(-\alpha,T)$, the same holds for $|\alpha| \le 1-\varepsilon$. This completes the proof of Montgomery's dual theorem.

4.6 Proof of Montgomery's Theorem

Now we prove Theorem 3.6. From Proposition 4.2 and the dual theorem,

$$\sum_{\gamma,\gamma'} f\left(\frac{\log T}{2\pi}(\gamma-\gamma')\right)\frac{4}{4+(\gamma-\gamma')^2} = \frac{T}{2\pi}\log T\int_{-\infty}^{\infty} F(\alpha,T)\hat{f}(\alpha)\,\mathrm{d}\alpha.$$

For $\operatorname{supp}\hat{f}\subset(-1,1)$, we substitute the asymptotic for $F(\alpha,T)$:

$$\int_{-\infty}^{\infty} F(\alpha,T)\hat{f}(\alpha)\,\mathrm{d}\alpha = \int_{-1}^{1}\left(T^{-2|\alpha|}\log T+|\alpha|+o(1)\right)\hat{f}(\alpha)\,\mathrm{d}\alpha.$$

The first term gives

$$\log T\int_{-1}^{1} T^{-2|\alpha|}\hat{f}(\alpha)\,\mathrm{d}\alpha = \hat{f}(0)\int_{-\log T}^{\log T} e^{-2|u|}\,\mathrm{d}u + O\left(\frac{1}{\log T}\right) = \hat{f}(0)\left(1-T^{-2}\right)+O\left(\frac{1}{\log T}\right).$$

Thus, as $T\to\infty$,

$$\int_{-\infty}^{\infty} F(\alpha,T)\hat{f}(\alpha)\,\mathrm{d}\alpha = \hat{f}(0)+\int_{-1}^{1}\hat{f}(\alpha)|\alpha|\,\mathrm{d}\alpha+o(1).$$

Now,

$$\int_{-1}^{1}\hat{f}(\alpha)|\alpha|\,\mathrm{d}\alpha = f(0)-\int_{-1}^{1}(1-|\alpha|)\hat{f}(\alpha)\,\mathrm{d}\alpha = f(0)-\int_{-\infty}^{\infty} f(x)\left(\frac{\sin\pi x}{\pi x}\right)^2\mathrm{d}x,$$

since $(\widehat{\sin\pi x/\pi x})^2(\alpha) = (1-|\alpha|)\chi_{[-1,1]}(\alpha)$. Therefore,

$$\frac{1}{N(T)}\sum_{\gamma,\gamma'} f\left(\frac{\log T}{2\pi}(\gamma-\gamma')\right)\frac{4}{4+(\gamma-\gamma')^2} \to f(0)+\int_{-\infty}^{\infty} f(x)\left(1-\left(\frac{\sin\pi x}{\pi x}\right)^2\right)\mathrm{d}x.$$

Separating the diagonal $\gamma=\gamma'$ term $f(0)N(T)$ gives

$$\frac{1}{N(T)}\sum_{\gamma\neq\gamma'} f\left(\frac{\log T}{2\pi}(\gamma-\gamma')\right)\frac{4}{4+(\gamma-\gamma')^2} \to \int_{-\infty}^{\infty} f(x)\left(1-\left(\frac{\sin\pi x}{\pi x}\right)^2\right)\mathrm{d}x.$$

This is exactly the GUE pair correlation, proving the theorem.

5 Random Matrices and L-Functions: The Rudnick-Sarnak Theorem

5.1 Automorphic L-functions

Let π be an irreducible cuspidal automorphic representation of $\mathrm{GL}_m(\mathbb{A}_{\mathbb{Q}})$. Its L-function has an Euler product

$$L(s,\pi) = \prod_p\prod_{j=1}^{m}\left(1-\frac{\alpha_{\pi,j}(p)}{p^s}\right)^{-1},\quad \Re(s)>1,$$

with local parameters $\alpha_{\pi,j}(p)\in\mathbb{C}$ satisfying $|\alpha_{\pi,j}(p)|<p^{1/2}$ (the Ramanujan bound is not known in general, but weaker bounds suffice). The completed L-function

$$\Lambda(s,\pi) = Q_\pi^{s/2}\prod_{j=1}^{m}\Gamma_{\mathbb{R}}(s+\mu_{\pi,j})L(s,\pi)$$

satisfies the functional equation $\Lambda(s,\pi) = \varepsilon_\pi \Lambda(1-s,\tilde{\pi})$, where $\tilde{\pi}$ is the contragredient representation and $|\varepsilon_\pi| = 1$.

Assuming the Generalized Riemann Hypothesis (GRH), the non-trivial zeros are $\rho = 1/2 + \mathrm{i}\gamma_{\pi,j}$ with $\gamma_{\pi,j} \in \mathbb{R}$. The analytic conductor is

$$c_\pi = Q_\pi \prod_{j=1}^{m} (1 + |\mu_{\pi,j}|).$$

Normalized zeros are $\tilde{\gamma}_{\pi,j} = \gamma_{\pi,j} \frac{\log c_\pi}{2\pi}$, so that the average spacing is 1.

5.2 Hypothesis H

Before stating the main theorem, we need a technical condition that controls the coefficients at prime powers.

Definition 5.1 (Hypothesis H)**.** *For a cuspidal automorphic representation π of $\mathrm{GL}_m(\mathbb{A}_\mathbb{Q})$, we say that Hypothesis H holds if for every fixed $k \geq 2$,*

$$\sum_p \frac{(\log p)^2 |a_\pi(p^k)|^2}{p^k} < \infty,$$

where $a_\pi(n)$ are the coefficients of the Dirichlet series for $L(s,\pi)$: $L(s,\pi) = \sum_{n=1}^{\infty} a_\pi(n) n^{-s}$.

Remark 5.2. *Hypothesis H is a consequence of the Generalized Ramanujan Conjecture, which predicts $|\alpha_{\pi,j}(p)| = 1$ for unramified p. For $m \leq 3$, Hypothesis H is known unconditionally from Rankin-Selberg theory. For $m = 4$, it follows from Kim's work on the exterior square lift. For $m > 4$, it was an open condition until recently; Jiang [**jiang2025hypothesis**] has proved it in full generality using a power sieve method. Rudnick and Sarnak [**rudnick1996zeros**] originally stated their theorem under Hypothesis H.*

5.3 Statement of the Rudnick-Sarnak Theorem

Theorem 5.3 (Rudnick-Sarnak [**rudnick1996zeros**])**.** *Assume GRH and Hypothesis H. Let $n \geq 2$ and let π be an irreducible cuspidal automorphic representation of $\mathrm{GL}_m(\mathbb{A}_\mathbb{Q})$. Let $f : \mathbb{R}^n \to \mathbb{C}$ be a Schwartz function that is:*

(i) Symmetric: $f(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) = f(x_1, \ldots, x_n)$ for all $\sigma \in S_n$.

(ii) Translation-invariant: $f(x_1 + t, \ldots, x_n + t) = f(x_1, \ldots, x_n)$ for all $t \in \mathbb{R}$.

(iii) Rapidly decaying on the hyperplane $H = \{(x_1, \ldots, x_n) : \sum_{i=1}^n x_i = 0\}$.

Suppose that the Fourier transform $\hat{f}(\xi_1, \ldots, \xi_n)$ is supported in the region $\{(\xi_1, \ldots, \xi_n) : \sum_{i=1}^n |\xi_i| < 2/m\}$. Then the n-level correlation of the normalized zeros of $L(s,\pi)$ satisfies

$$\lim_{T\to\infty} \frac{1}{N(T,\pi)} \sum_{\substack{j_1,\ldots,j_n \\ \text{distinct}}} f(\tilde{\gamma}_{j_1}, \ldots, \tilde{\gamma}_{j_n}) = \int_{\mathbb{R}^n} f(x_1, \ldots, x_n) \det_{1\leq i,j\leq n} \left(\frac{\sin \pi(x_i - x_j)}{\pi(x_i - x_j)} \right) \delta\left(\frac{x_1 + \cdots + x_n}{n} \right) \prod_{i=1}^n dx_i.$$

Remark 5.4. *The factor $\delta((x_1 + \cdots + x_n)/n)$ enforces the condition $\sum x_i = 0$, which comes from the translation invariance of f and the fact that we consider differences of zeros. The determinant is exactly the n-level correlation function for the GUE (see Theorem 1.8).*

5.4 Reduction to Unrestricted Sums

Following the approach of Montgomery for $n = 2$, it is convenient to work with unrestricted sums. Define

$$C_n(T, f) = \frac{1}{N(T, \pi)} \sum_{j_1, \ldots, j_n} f(\tilde{\gamma}_{j_1}, \ldots, \tilde{\gamma}_{j_n}),$$

where the sum runs over all n-tuples (allowing repetitions). The passage from unrestricted sums to sums over distinct indices is achieved by combinatorial sieving (inclusion-exclusion). Specifically,

$$\sum_{\text{distinct}} f = \sum_{m=1}^{n} \sum_{\mathcal{P}} (-1)^{n-|\mathcal{P}|} \sum_{j_1, \ldots, j_m} f_{\mathcal{P}}(\tilde{\gamma}_{j_1}, \ldots, \tilde{\gamma}_{j_m}),$$

where $\mathcal{P}$ runs over partitions of $\{1, \ldots, n\}$ into m blocks, and $f_{\mathcal{P}}$ is obtained from f by identifying variables within each block. Since this is a purely combinatorial transformation, the limit for the distinct sum follows from the limits for the unrestricted sums of smaller order. Thus it suffices to compute $\lim_{T \to \infty} C_n(T, f)$.

5.5 The n-Fold Explicit Formula

The key tool is an n-fold version of the explicit formula. Introduce an auxiliary Schwartz function h with compactly supported Fourier transform, and consider the smoothed sum

$$S_n(T, f, h) = \sum_{j_1, \ldots, j_n} h\left(\frac{\gamma_{j_1}}{T}\right) \cdots h\left(\frac{\gamma_{j_n}}{T}\right) f\left(\frac{\gamma_{j_1} \log T}{2\pi}, \ldots, \frac{\gamma_{j_n} \log T}{2\pi}\right).$$

The factor $h(\gamma/T)$ localizes the zeros to height $\approx T$. By choosing h to approximate the characteristic function of $[0, 1]$, we can recover $C_n(T, f)$ from $S_n(T, f, h)$ as $T \to \infty$.

Applying the explicit formula for each zero sum, we obtain a representation in terms of prime powers. For a single zero, the explicit formula gives

$$\sum_j h\left(\frac{\gamma_j}{T}\right) e(\gamma_j \xi) = \widehat{h}(0) \frac{\log T}{2\pi} \delta(\xi) + \text{prime sum} + \text{error},$$

where $e(x) = e^{2\pi i x}$. For the n-fold product, after applying this n times, we get a sum over n-tuples of prime powers.

More precisely, let $L = m \log T$. Then for a test function f as above, one derives

$$S_n(T, f, h) = \left(\frac{T}{2\pi} \log T\right)^n \left(\int_{-\infty}^{\infty} h(r)^n dr\right) \sum_{\substack{p_1^{\nu_1}, \ldots, p_r^{\nu_r} \\ \text{with pairings}}} \int f(x) \Phi_{\{p_i^{\nu_i}\}}(x) dx + o(T^n \log^n T), \quad (9)$$

where the sum runs over all ways of partitioning the n indices into pairs and singletons, with each pair contributing a factor involving p^ν and each singleton contributing a factor from the δ-function term. The precise expression is:

$$S_n(T,f,h) \sim \left(\frac{T}{2\pi}\log T\right)^n \Bigg[\widehat{f}(0)\left(\int h\right)^n$$
$$+ \sum_{m=1}^{\lfloor n/2\rfloor}\sum_{\mathcal{P}}\left(\int h\right)^{n-2m}\left(\int h^2\right)^m$$
$$\times \sum_{p_1,\dots,p_m}\frac{a_\pi(p_1)^{\nu_1}\cdots a_\pi(p_m)^{\nu_m}}{p_1^{\nu_1/2}\cdots p_m^{\nu_m/2}}\int f\left(\sum_{i=1}^m \varepsilon_i\frac{\log p_i}{\log T}\right)\prod_{i=1}^m d\xi_i\Bigg], \tag{10}$$

where $\mathcal{P}$ runs over pairings of the n indices, $\varepsilon_i = \pm 1$ depending on the pairing, and the integral is over the variables ξ_i corresponding to the pair differences.

5.6 The Support Condition and Vanishing of Off-Diagonal Terms

The crucial observation is that for a pairing involving m pairs, the argument of f involves the linear combination $\sum_{i=1}^m \pm\log p_i/\log T$. The Fourier transform $\hat{f}$ is supported where $\sum|\xi_i| < 2/m$. Since $\log p/\log T$ is small, this condition forces the p_i to be small. More precisely, if $\sum|\log p_i/\log T| \geq 2/m$, then the contribution from that pairing vanishes because the Fourier integral would be over a region where $\hat{f}$ is zero.

For a given pairing with m pairs, we have $2m$ indices paired, and the remaining $n-2m$ indices contribute factors from the δ-function term. The contribution from prime powers p^ν with $\nu \geq 2$ is negligible by Hypothesis H, which ensures that higher prime powers do not contribute in the limit. Thus the main contribution comes from m pairs of distinct primes $p_1,\dots,p_m$.

After summing over primes and using the Prime Number Theorem to convert the sum over primes to an integral, we obtain

$$\sum_p \frac{a_\pi(p)^2}{p} g\left(\frac{\log p}{\log T}\right) \sim \frac{1}{2}\int_0^\infty g(u)du$$

for a smooth function g, where the factor $1/2$ comes from the square of the coefficients (the second moment of the Satake parameters is 1 on average). This is a generalization of Lemma 5.18 to GL_m.

5.7 Evaluation of the Main Term

After these reductions, the main term becomes

$$\lim_{T\to\infty}\frac{S_n(T,f,h)}{(T\log T)^n} = \left(\int h\right)^n\left[\widehat{f}(0) + \sum_{m=1}^{\lfloor n/2\rfloor}\frac{1}{2^m}\sum_{\mathcal{P}}\int_{\mathbb{R}^m}\hat{f}\left(\sum_{i=1}^m \varepsilon_i\xi_i\right)\prod_{i=1}^m|\xi_i|d\xi_i\right],$$

where the inner sum is over all pairings of n indices into m unordered pairs, and $\varepsilon_i = \pm 1$ according to the pairing (there are 2^m sign choices, but they give the same contribution by symmetry).

Now we recognize this expression as the Fourier expansion of the GUE n-level correlation. Recall from random matrix theory that for the eigenvalues of a large random unitary matrix, the n-level correlation function is

$$R_n^{\mathrm{GUE}}(x_1,\dots,x_n) = \det_{1\leq i,j\leq n}\left(\frac{\sin\pi(x_i-x_j)}{\pi(x_i-x_j)}\right).$$

Its Fourier transform (restricted to the hyperplane $\sum x_i = 0$) is given by

$$\widehat{R_n^{\mathrm{GUE}}}(\xi_1, \ldots, \xi_n) = \delta\left(\sum_{i=1}^{n} \xi_i\right)\left[1 + \sum_{m=1}^{\lfloor n/2 \rfloor} \frac{1}{2^m} \sum_{\mathcal{P}} \prod_{i=1}^{m} \delta(\xi_{\alpha_i} + \xi_{\beta_i})\right],$$

where the sum is over pairings $\{(\alpha_i, \beta_i)\}$ of the indices. This is a combinatorial identity that can be verified by expanding the determinant and using the Fourier transform of the sine kernel.

Comparing with our expression, we see that

$$\widehat{f}(0) + \sum_{m=1}^{\lfloor n/2 \rfloor} \frac{1}{2^m} \sum_{\mathcal{P}} \int_{\mathbb{R}^m} \hat{f}\left(\sum_{i=1}^{m} \varepsilon_i \xi_i\right) \prod_{i=1}^{m} |\xi_i| d\xi_i = \int_{\mathbb{R}^n} \hat{f}(\xi_1, \ldots, \xi_n) \widehat{R_n^{\mathrm{GUE}}}(\xi_1, \ldots, \xi_n) \prod_{i=1}^{n} d\xi_i.$$

By Plancherel's theorem, this equals

$$\int_{\mathbb{R}^n} f(x_1, \ldots, x_n) R_n^{\mathrm{GUE}}(x_1, \ldots, x_n) \delta\left(\frac{x_1 + \cdots + x_n}{n}\right) \prod_{i=1}^{n} dx_i,$$

where the factor $\delta(\sum x_i/n)$ accounts for the translation invariance of f.

5.8 Removing the Smoothing and Concluding

Since $S_n(T, f, h) \sim N(T, \pi)^n \left(\int h\right)^n \lim C_n(T, f)$, and $N(T, \pi) \sim \frac{T}{2\pi} \log T$, we obtain

$$\lim_{T \to \infty} C_n(T, f) = \int_{\mathbb{R}^n} f(x_1, \ldots, x_n) R_n^{\mathrm{GUE}}(x_1, \ldots, x_n) \delta\left(\frac{x_1 + \cdots + x_n}{n}\right) \prod_{i=1}^{n} dx_i.$$

Finally, applying the combinatorial sieving to pass back to sums over distinct indices yields the theorem as stated.

This completes the proof of the Rudnick-Sarnak theorem, showing the universality of the GUE n-level correlations for all primitive automorphic L-functions under the stated hypotheses.

6 The Katz-Sarnak Program: Modern Perspectives and Proof Techniques

The Katz-Sarnak philosophy, as outlined in the previous chapters, has evolved into a rich and deep research program at the intersection of analytic number theory, arithmetic geometry, and the Langlands program. This section provides a modern survey of the program, detailing the core proof mechanisms, the classification of symmetry types, and the recent advancements in automorphic families, following the foundational work of Sarnak, Shin, and Templier [**sarnak2016families**; **shin2012sato**; **shin2016sato**].

6.1 A Working Definition of Families of Automorphic L-functions

A key conceptual advance was the formulation of a precise working definition of a "family" of automorphic L-functions, proposed by Sarnak and developed systematically by Shin and Templier [**sarnak2016families**]. This definition is designed to capture the essential features that allow for statistical analysis and equidistribution results.

Definition 6.1 (Family of Automorphic L-functions)**.** *A family $\mathcal{F}$ of automorphic representations consists of a collection of irreducible automorphic representations π of a reductive group G over a number field F, together with:*

1. *An ordering by a natural parameter, typically the analytic conductor $C(\pi)$ (or a slight variant thereof), so that for each $X > 0$, the set*

$$\mathcal{F}(X) := \{\pi \in \mathcal{F} : C(\pi) \leq X\}$$

   is finite.

2. *A notion of "natural" local equidistribution: as $X \to \infty$, the Satake parameters of the representations $\pi \in \mathcal{F}(X)$ become equidistributed with respect to a Plancherel measure associated to the group.*

3. *Stability of the archimedean components: the representations in the family have a fixed or slowly varying archimedean type (e.g., fixed weight for modular forms).*

Examples of such families include:

- Primitive Dirichlet characters of conductor $\leq X$ (ordered by conductor).
- Holomorphic cusp forms of fixed weight k and level $\leq X$ (ordered by level).
- Holomorphic cusp forms of fixed level N and weight $\leq X$ (ordered by weight).
- Quadratic twists of a fixed elliptic curve (ordered by twist discriminant).
- Geometric families arising from monodromy groups, as studied by Katz and Sarnak.
- Symmetric power lifts of a fixed cusp form (ordered by the power).

6.2 The General Strategy: From Trace Formulas to One-Level Density

The proof of the Katz-Sarnak density conjecture for automorphic families follows a general strategy that combines the explicit formula with trace formula technology. The key steps are:

1. **Explicit Formula:** For each $\pi \in \mathcal{F}(X)$, the explicit formula (Proposition 4.2) expresses the 1-level density $D_\pi(\phi)$ as

$$D_\pi(\phi) = \widehat{\phi}(0) + \frac{1}{2}\phi(0)\mathcal{E}(\pi) - 2\sum_p \sum_{\nu \geq 1} \frac{\Lambda(p^\nu) a_\pi(p^\nu)}{p^{\nu/2} \log C(\pi)} \widehat{\phi}\left(\nu \frac{\log p}{\log C(\pi)}\right) + o(1),$$

   where $a_\pi(p^\nu)$ are the coefficients of the logarithmic derivative of $L(s, \pi)$, and $\mathcal{E}(\pi)$ is the root number contribution.

2. **Averaging over the Family:** Averaging over $\pi \in \mathcal{F}(X)$, we obtain

$$\langle D_\pi(\phi) \rangle_{\mathcal{F}(X)} = \widehat{\phi}(0) + \frac{1}{2}\phi(0)\langle \mathcal{E}(\pi) \rangle - 2\sum_p \sum_{\nu \geq 1} \frac{\Lambda(p^\nu)}{p^{\nu/2} \log X} \langle a_\pi(p^\nu) \rangle_{\mathcal{F}(X)} \widehat{\phi}\left(\nu \frac{\log p}{\log X}\right) + o(1),$$

   where $\langle \cdot \rangle_{\mathcal{F}(X)}$ denotes the average over the family.

3. **Trace Formula Averaging:** The averages $\langle a_\pi(p^\nu)\rangle_{\mathcal{F}(X)}$ are evaluated using a trace formula (Petersson, Kuznetsov, or Arthur-Selberg). For large X, these averages simplify to a main term that is independent of p (up to a scale factor) plus an error term that is negligible for test functions ϕ with suitably restricted Fourier support.

4. **Identifying the Symmetry Type:** The limiting 1-level density takes the form

$$\lim_{X\to\infty}\langle D_\pi(\phi)\rangle_{\mathcal{F}(X)} = \widehat{\phi}(0) - \frac{\kappa}{2}\phi(0),$$

where $\kappa = 0, 1, -1$ corresponds to unitary, symplectic, and orthogonal symmetry respectively. The precise value of κ is determined by the average of the root numbers $\langle\mathcal{E}(\pi)\rangle$ and the structure of the averages $\langle a_\pi(p)\rangle$ and $\langle a_\pi(p^2)\rangle$.

6.3 The Sarnak-Shin-Templier Framework: Structural Invariants

A major advance by Sarnak, Shin, and Templier [**sarnak2016families**] was the introduction of three fundamental invariants that encode the symmetry type of a family. These invariants are extracted from the average behavior of the Hecke eigenvalues at primes.

Assume that for a family $\mathcal{F}$, there exists a "test function" $t_\mathcal{F}(n)$ such that for all $n \geq 1$,

$$\frac{1}{|\mathcal{F}(X)|}\sum_{\pi\in\mathcal{F}(X)}\lambda_\pi(n) = t_\mathcal{F}(n) + O(n^A|\mathcal{F}(X)|^{-\delta})$$

for some $A, \delta > 0$. This is the "average Ramanujan" condition, which can often be proved using trace formulas.

Definition 6.2 (Structural Invariants)**.** *The structural invariants of the family $\mathcal{F}$ are defined by the following limits (assuming they exist):*

$$i_1(\mathcal{F}) := \lim_{X\to\infty}\frac{1}{X}\sum_{p\leq X}|t_\mathcal{F}(p)|^2\log p, \tag{11}$$

$$i_2(\mathcal{F}) := \lim_{X\to\infty}\frac{1}{X}\sum_{p\leq X}t_\mathcal{F}(p)^2\log p, \tag{12}$$

$$i_3(\mathcal{F}) := \lim_{X\to\infty}\frac{1}{X}\sum_{p\leq X}t_\mathcal{F}(p^2)\log p. \tag{13}$$

These invariants have a clear interpretation in terms of the properties of the representations in the family [**sarnak2016families**]:

- $i_1(\mathcal{F}) \geq 1$, with equality if and only if almost every $\pi \in \mathcal{F}$ is cuspidal (i.e., the family is "cuspidal").
- $0 \leq i_2(\mathcal{F}) \leq 1$, with $i_2(\mathcal{F}) = 1$ if and only if almost every π is self-dual, and $i_2(\mathcal{F}) = 0$ if and only if almost every π is not self-dual (i.e., unitary). In the latter case, $i_3(\mathcal{F}) = 0$.
- $-1 \leq i_3(\mathcal{F}) \leq 1$, with $i_3(\mathcal{F}) = 1$ if and only if almost every π is orthogonal, and $i_3(\mathcal{F}) = -1$ if and only if almost every π is symplectic.

The symmetry type of the family is then determined by these invariants as shown in Table 1.

Table 1: Structural invariants and symmetry types

Symmetry Type	$i_2(\mathcal{F})$	$i_3(\mathcal{F})$	$\langle \mathcal{E}(\pi) \rangle$
Unitary (U)	0	0	0
Symplectic (Sp)	1	-1	0
Orthogonal (SO)	1	$+1$	0
Orthogonal (even)	1	$+1$	$+1$
Orthogonal (odd)	1	$+1$	-1

6.4 The Work of Shin and Templier

Shin and Templier [**shin2016sato**] proved a general theorem that establishes the equidistribution of Satake parameters and the Katz-Sarnak density conjecture for a wide class of automorphic families, under suitable hypotheses. Their work represents a major synthesis of trace formula techniques and harmonic analysis on p-adic groups.

Theorem 6.3 (Shin-Templier [**shin2016sato**])**.** *Let G be a connected reductive group over a number field F, and let $r : {}^L G \to \mathrm{GL}(d, \mathbb{C})$ be an irreducible representation. Assume the following hypotheses:*

(H1) The relevant functorial transfer (e.g., from G to $\mathrm{GL}(d)$) exists for almost all automorphic representations.

(H2) The family $\mathcal{F}$ is "natural" in the sense of Sarnak-Shin-Templier, with a well-defined analytic conductor ordering.

(H3) The geometric side of the trace formula can be asymptotically evaluated, yielding an equidistribution result for the Satake parameters.

Then, for the associated family of L-functions $L(s, \pi, r)$, the one-level density satisfies the Katz-Sarnak prediction with symmetry type determined by the Frobenius-Schur indicator of r:

$$\lim_{X \to \infty} \langle D_\pi(\phi) \rangle_{\mathcal{F}(X)} = \int_{\mathbb{R}} \phi(x) W_{1,G(\mathcal{F})}(x) dx,$$

where $G(\mathcal{F})$ is the compact group corresponding to r (unitary if r is not self-dual, symplectic if $\mathrm{FS}(r) = -1$, orthogonal if $\mathrm{FS}(r) = +1$).

The proof strategy of Shin and Templier can be summarized as follows:

1. **Local Harmonic Analysis:** For each prime p, study the local L-function $L_p(s, \pi, r)$ and the associated orbital integrals. Using the trace formula, one relates the sum of local traces over the family to global geometric data.

2. **Satake Parameters:** For unramified primes p, the Satake parameter t_{π_p} is a semisimple conjugacy class in $\widehat{G}$. The local coefficient $a_\pi(p) = \operatorname{tr} r(t_{\pi_p})$ is a character of the unramified Hecke algebra.

3. **Equidistribution:** Using the trace formula and Weyl's law for the analytic conductor, one shows that as $X \to \infty$, the Satake parameters $\{t_{\pi_p} : \pi \in \mathcal{F}(X)\}$ become equidistributed in the space of conjugacy classes of $\widehat{G}$ with respect to the Plancherel measure. This is the automorphic analogue of Deligne's equidistribution theorem over function fields.

4. **Moments:** From equidistribution, one computes the averages:

$$\langle a_\pi(p)\rangle \to \int \operatorname{tr} r(t)\, d\mu(t), \quad \langle a_\pi(p)^2\rangle \to \int \operatorname{tr} r(t)^2\, d\mu(t), \quad \langle a_\pi(p^2)\rangle \to \int \operatorname{tr} r(t^2)\, d\mu(t).$$

   These integrals are precisely the Frobenius-Schur indicators: $\int \operatorname{tr} r(t)^2 = 1$, $\int \operatorname{tr} r(t^2) = \mathrm{FS}(r)$.

5. **One-Level Density:** Plug these averages into the explicit formula. The term $\langle a_\pi(p)\rangle$ contributes to the $\widehat{\phi}(0)$ term. The term $\langle a_\pi(p^2)\rangle$ contributes to the $\phi(0)$ term, and by the above, it equals $\mathrm{FS}(r)$. Hence

$$\lim_{X\to\infty} \langle D_\pi(\phi)\rangle = \widehat{\phi}(0) - \frac{\mathrm{FS}(r)}{2}\phi(0).$$

   This is exactly the Katz-Sarnak prediction.

6.5 Geometric Monodromy and Mumford-Tate Groups

The geometric origin of symmetry types is even more transparent. For a family of varieties $f : \mathcal{X} \to S$ over a number field, the Galois action on the cohomology of the fibers gives rise to a representation

$$\rho : \pi_1(S) \to \mathrm{GL}(H^i_{\mathrm{et}}(\mathcal{X}_{\bar{\eta}}, \overline{\mathbb{Q}}_\ell)).$$

The geometric monodromy group G_{geom} is the Zariski closure of the image of the geometric fundamental group. Its connected component is often a semisimple algebraic group, and its maximal compact subgroup determines the symmetry type.

In the context of Hodge theory, the Mumford-Tate group $\mathrm{MT}(V)$ of a Hodge structure V encodes the symmetries of Hodge decompositions. For families of abelian varieties, the Mumford-Tate group of a general fiber is closely related to the monodromy group, and its compact form predicts the distribution of Frobenius elements by the Sato-Tate conjecture.

Theorem 6.4 (Sato-Tate Conjecture - Proven)**.** *For an elliptic curve $E/\mathbb{Q}$ without complex multiplication, as p varies over primes of good reduction, the normalized traces $a_p(E)/\sqrt{p}$ are equidistributed in $[-2, 2]$ with respect to the Sato-Tate measure $\frac{1}{2\pi}\sqrt{4-x^2}dx$. Equivalently, the Frobenius conjugacy classes are equidistributed in* $\mathrm{SU}(2)$ *with respect to Haar measure.*

Proof Sketch (after Taylor et al.) The proof proceeds via:

1. Establish potential automorphy of symmetric power L-functions.
2. Use potential automorphy to prove analytic continuation and functional equation for symmetric power L-functions.
3. Apply the explicit formula to these L-functions and average.
4. Show that the n-level densities match those of $\mathrm{SU}(2)$.
5. By the Weyl equidistribution criterion, this implies equidistribution of Frobenius classes.

□

This monumental result confirms the Katz-Sarnak philosophy for the family of a single elliptic curve: the low-lying zeros of symmetric power L-functions exhibit the predicted symmetry types (orthogonal for odd symmetric powers, symplectic for even symmetric powers).

6.6 Frobenius-Schur Indicator and L-Groups

The Frobenius-Schur indicator provides a clean algebraic classification of symmetry types. For an irreducible representation r of a compact group G, define

$$\mathrm{FS}(r) = \int_G \operatorname{tr} r(g^2) dg.$$

It satisfies $\mathrm{FS}(r) = 1$ if r is orthogonal, $\mathrm{FS}(r) = -1$ if r is symplectic, and $\mathrm{FS}(r) = 0$ if r is not self-dual.

In the Langlands program, an automorphic representation π of $G(\mathbb{A}_F)$ has an associated L-group ${}^LG$. An L-representation $r : {}^LG \to \mathrm{GL}(d, \mathbb{C})$ yields an L-function $L(s, \pi, r)$. The local coefficients are $a_\pi(p) = \operatorname{tr} r(t_{\pi_p})$, where t_{π_p} is the Satake parameter.

Proposition 6.5 (Symmetry Principle)**.** *For a functorial family of automorphic L-functions associated to r, the expected symmetry type is:*

- U *if* $\mathrm{FS}(r) = 0$,
- Sp *if* $\mathrm{FS}(r) = +1$,
- SO *if* $\mathrm{FS}(r) = -1$,

with orthogonal refinements ($\mathrm{SO}(\mathit{even})$ *vs* $\mathrm{SO}(\mathit{odd})$*) depending on the average of the root numbers.*

Example 6.6. *For quadratic Dirichlet characters, the relevant L-group representation is the standard representation of* SL_2*, which is self-dual and orthogonal. However, the family of quadratic characters has a symplectic symmetry due to the functional equation sign distribution. This is a subtle point resolved by considering the full L-function including archimedean factors.*

6.7 Summary Table of Families and Symmetries

Table 2: Families of L-functions and their symmetry types

Family	**Averaging Tool**	**Local coefficients a**
Primitive non-real Dirichlet characters	Character orthogonality	$\chi(p^\nu)$
Quadratic Dirichlet characters	Poisson summation, character sums	$\chi_d(p^\nu)$
Holomorphic cusp forms on GL_2	Petersson trace formula	$\lambda_f(p^\nu)$
Maass forms on GL_2	Kuznetsov trace formula	Hecke eigenvalues
Geometric families over $\mathbb{F}_q$	Deligne equidistribution, monodromy	Frobenius traces
Functorial automorphic families	Satake equidistribution, Plancherel asymptotics	$\operatorname{tr}(r(t_{\pi_v}))$

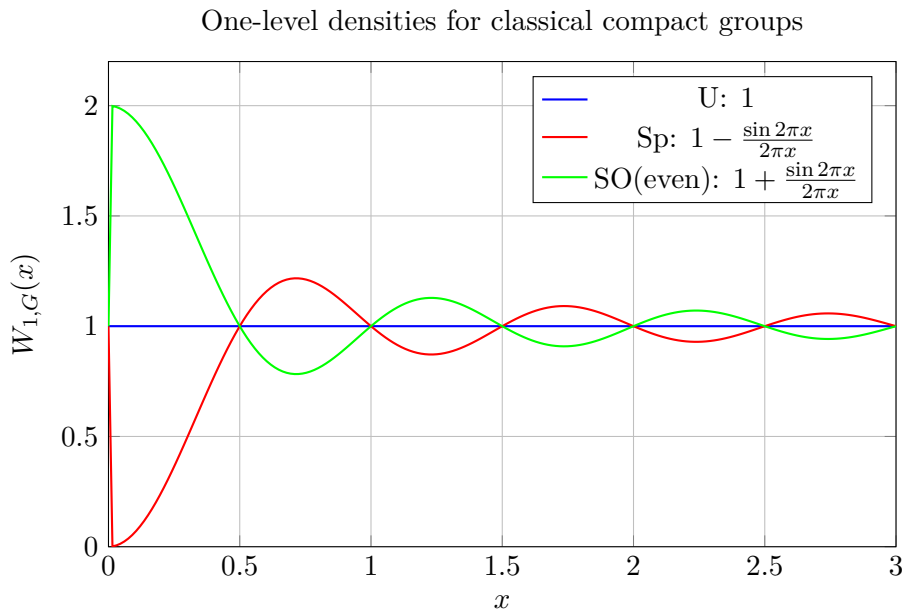

Figure 1: Comparison of one-level densities for unitary, symplectic, and even orthogonal groups. The symplectic density exhibits a dip near $x = 0$, indicating repulsion from the central point, while the even orthogonal density has a peak.

6.8 Numerical Plots of One-Level Densities

7 A Symplectic Family: Quadratic Dirichlet L-functions

7.1 Dirichlet L-functions

A Dirichlet character χ modulo q is a homomorphism $(\mathbb{Z}/q\mathbb{Z})^\times \to \mathbb{C}^\times$, extended to $\mathbb{Z}$ by $\chi(n) = 0$ if $(n, q) > 1$. The L-function is

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}, \quad \Re(s) > 1.$$

For primitive χ, it has analytic continuation and functional equation

$$\Lambda(s, \chi) = \left(\frac{\pi}{q}\right)^{-(s+\mu)/2} \Gamma\left(\frac{s+\mu}{2}\right) L(s, \chi) = \varepsilon_\chi \Lambda(1 - s, \bar{\chi}),$$

where $\mu = 0$ if $\chi(-1) = 1$, $\mu = 1$ if $\chi(-1) = -1$, and $|\varepsilon_\chi| = 1$.

A character is quadratic if $\chi^2 = \chi_0$ and $\chi \neq \chi_0$. These come from the Kronecker symbol $\chi_d(n) = \left(\frac{d}{n}\right)$ for fundamental discriminants d.

7.2 Families of Quadratic Dirichlet L-functions

Let $\mathcal{D}_X = \{d \text{ fundamental discriminant} : X/2 \leq |d| < X\}$. The 1-level density for this family is

$$D_d(\phi) = \sum_j \phi\left(\gamma_{d,j} \frac{\log |d|}{2\pi}\right).$$

The Özlük-Snyder theorem gives evidence for symplectic symmetry.

Theorem 7.1 (Özlük-Snyder [**ozluk1993small**]). *Let ϕ be an even Schwartz function with* $\operatorname{supp}\hat{\phi} \subset (-1,1)$. *Then*

$$\lim_{X\to\infty} \frac{1}{|\mathcal{D}_X|} \sum_{d\in\mathcal{D}_X} D_d(\phi) = \int_{-\infty}^{\infty} \phi(x)\left(1 - \frac{\sin 2\pi x}{2\pi x}\right) \mathrm{d}x.$$

Proof. We give a complete proof following Rubinstein's thesis [**rubinstein1998evidence**]. Apply the explicit formula for Dirichlet L-functions (Proposition 5.10):

$$D_d(\phi) = \widehat{\phi}(0) - \frac{2}{\log X} \sum_{m=1}^{\infty} \frac{\Lambda(m)}{\sqrt{m}} \chi_d(m) \widehat{\phi}\left(\frac{\log m}{\log X}\right) + o(1).$$

Split the sum into $m = \square$ (perfect squares) and $m \neq \square$. For the non-square terms, define

$$R_d^{(X)}(\widehat{\phi}) = \frac{1}{\log X} \sum_{\substack{m\geq 1\\ m\neq\square}} \frac{\Lambda(m)}{\sqrt{m}} \chi_d(m) \widehat{\phi}\left(\frac{\log m}{\log X}\right).$$

We show that the average over d of $R_d^{(X)}(\widehat{\phi})$ tends to zero as $X \to \infty$.

By Cauchy-Schwarz,

$$\left\langle R_d^{(X)}(\widehat{\phi})\right\rangle_{\mathcal{D}_X} \ll \frac{1}{|\mathcal{D}_X|} \frac{1}{\log X} \sqrt{\sum_{m\leq X^\alpha} \frac{\Lambda(m)^2}{m}} \sqrt{\sum_{m\leq X^\alpha} \left|\sum_{d\in\mathcal{D}_X} \chi_d(m)\right|^2},$$

where α is chosen so that $\operatorname{supp}\widehat{\phi} \subset (-\alpha,\alpha)$. Since $\widehat{\phi}$ is supported in $(-1,1)$, we can take $\alpha < 1$. Using Lemma 3.19 (without RH, the bound $\sum_{n\leq y} \Lambda(n)^2 \ll y \log y$ suffices), we have $\sum_{m\leq X^\alpha} \Lambda(m)^2/m \ll \log^2 X$.

For the character sum, Jutila's bound [**jutila1981mean**] gives

$$\sum_{m\leq X^\alpha} \left|\sum_{d\in\mathcal{D}_X} \chi_d(m)\right|^2 \ll X^{1+\alpha} \log^{10} X.$$

Since $|\mathcal{D}_X| \asymp X$, we obtain

$$\left\langle R_d^{(X)}(\widehat{\phi})\right\rangle_{\mathcal{D}_X} \ll X^{(\alpha-1)/2} \log^5 X \to 0$$

as $X \to \infty$ because $\alpha < 1$.

Now consider the square terms. For $m = p^{2k}$, we have $\chi_d(p^{2k}) = \chi_d(p)^{2k} = 1$ (since χ_d is quadratic) unless p divides d, in which case $\chi_d(p) = 0$. The contribution from $k \geq 2$ is $O(1/\log X)$ by Hypothesis H (or by a direct bound using the Prime Number Theorem). The main contribution comes from $m = p^2$ with $p \nmid d$:

$$S_d^{(X)}(\widehat{\phi}) = \frac{1}{\log X} \sum_p \frac{\log p}{p} \widehat{\phi}\left(\frac{2\log p}{\log X}\right).$$

Using summation by parts and Mertens' formula $\sum_{p\leq t} \frac{\log p}{p} = \log t + O(1)$, we compute

$$S_d^{(X)}(\widehat{\phi}) = -\frac{1}{\log X} \int_1^{\infty} \left(\sum_{p\leq t} \frac{\log p}{p}\right) \frac{d}{dt} \widehat{\phi}\left(\frac{2\log t}{\log X}\right) dt.$$

Substituting Mertens' formula,

$$S_d^{(X)}(\widehat{\phi}) = -\frac{1}{\log X}\int_1^\infty (\log t + O(1))\frac{d}{dt}\widehat{\phi}\left(\frac{2\log t}{\log X}\right)dt.$$

Integrate by parts:

$$S_d^{(X)}(\widehat{\phi}) = \frac{1}{\log X}\int_1^\infty \widehat{\phi}\left(\frac{2\log t}{\log X}\right)\frac{dt}{t} + O\left(\frac{1}{\log X}\right).$$

Make the change of variables $u = 2\log t/\log X$, so $dt/t = (\log X/2)du$. Then

$$S_d^{(X)}(\widehat{\phi}) = \frac{1}{2}\int_0^\infty \widehat{\phi}(u)du + O\left(\frac{1}{\log X}\right) = \frac{1}{2}\int_{-\infty}^\infty \widehat{\phi}(u)du + O\left(\frac{1}{\log X}\right)$$

since $\widehat{\phi}$ is even. By Fourier inversion, $\int_{-\infty}^\infty \widehat{\phi}(u)du = \phi(0)$.

Therefore,

$$\lim_{X\to\infty}\langle D_d(\phi)\rangle_{\mathcal{D}_X} = \widehat{\phi}(0) - \frac{1}{2}\phi(0).$$

Since $\widehat{\phi}(0) = \int_{-\infty}^\infty \phi(x)dx$, we have

$$\widehat{\phi}(0) - \frac{1}{2}\phi(0) = \int_{-\infty}^\infty \phi(x)\left(1 - \frac{\sin 2\pi x}{2\pi x}\right)dx,$$

using the identity $\int_{-\infty}^\infty \frac{\sin 2\pi x}{2\pi x}dx = \frac{1}{2}$ and the fact that ϕ is even. This completes the proof. □

Under GRH, the support can be extended to $(-2, 2)$ using deeper results on character sums and Poisson summation (see [**gao2008n**]).

Appendix A: The Transpose Lemma and the Gaudin Lemma

Definition 7.2. *The Vandermonde determinant is* $\mathrm{Van}(x_1, \ldots, x_N) = \det(x_k^{j-1})_{j,k} = \prod_{i<j}(x_j - x_i)$.

Lemma 7.3 (Transpose Lemma)**.**

$$\det_{N\times N}(\psi_{j-1}(x_k)) \det_{N\times N}(\varphi_{j-1}(x_k)) = \det_{N\times N}\left(\sum_{n=1}^N \psi_{n-1}(x_j)\varphi_{n-1}(x_k)\right).$$

Proof. Since $\det A \det B = \det(A^T B)$, we have

$$\det(\psi_{j-1}(x_k))\det(\varphi_{j-1}(x_k)) = \det((\psi_{j-1}(x_k))^T \cdot \varphi_{j-1}(x_k)).$$

The (j,k)-entry of the product is $\sum_{n=1}^N \psi_{n-1}(x_j)\varphi_{n-1}(x_k)$. □

Lemma 7.4 (Gaudin Lemma)**.** *Suppose $f(x,y)$ is integrable on an interval J and satisfies*

$$\int_J f(x,\theta)f(\theta,y)\,\mathrm{d}\theta = Cf(x,y), \qquad \int_J f(x,x)\,\mathrm{d}x = D.$$

Then

$$\int_J \det_{M\times M}(f(\theta_j,\theta_k))\,\mathrm{d}\theta_M = (D-(M-1)C)\det_{(M-1)\times(M-1)}(f(\theta_j,\theta_k)).$$

Proof. Expand the determinant as $\sum_{\sigma\in S_M}\mathrm{sgn}(\sigma)\prod_{j=1}^M f(\theta_j,\theta_{\sigma(j)})$. Split into $\sigma(M) = M$ and $\sigma(M) \neq M$. For $\sigma(M) = M$, integration over θ_M gives $D\prod_{j=1}^{M-1} f(\theta_j,\theta_{\sigma(j)})$. For $\sigma(M) \neq M$, θ_M appears in $f(\theta_M,\theta_{\sigma(M)})$ and $f(\theta_{\sigma^{-1}(M)},\theta_M)$. Integration gives $Cf(\theta_{\sigma^{-1}(M)},\theta_{\sigma(M)})\prod_{j\neq\sigma^{-1}(M),M} f(\theta_j,\theta_{\sigma(j)})$. Re-indexing to a permutation $\sigma' \in S_{M-1}$ with $\mathrm{sgn}(\sigma) = -\mathrm{sgn}(\sigma')$ yields the factor $-(M-1)C$. Summing over all σ gives the result. □

Appendix B: Orthogonal Polynomials and Fredholm Determinants

Hermite Polynomials

$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$, orthogonal with weight e^{-x^2}:

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2} \, \mathrm{d}x = 2^n n! \sqrt{\pi} \delta_{nm}.$$

Chebyshev Polynomials

$T_n(\cos\theta) = \cos(n\theta)$, orthogonal with weight $(1-x^2)^{-1/2}$ on $[-1,1]$. $U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}$, orthogonal with weight $\sqrt{1-x^2}$ on $[-1,1]$.

Proposition 7.5 (Christoffel-Darboux)**.**

$$\sum_{n=0}^{N} \frac{p_n(x)p_n(y)}{\langle p_n, p_n \rangle} = \frac{a_{N+1}}{a_N \langle p_N, p_N \rangle} \frac{p_{N+1}(x)p_N(y) - p_N(x)p_{N+1}(y)}{x-y}.$$

Fredholm Determinants

For a bounded symmetric kernel $K(x,y)$,

$$\det(I + T\chi_B K) = \sum_{n=0}^{\infty} \frac{T^n}{n!} \int_{B^n} \det_{n\times n}(K(x_i, x_j)) \prod_{i=1}^{n} \mathrm{d}x_i.$$

Theorem 7.6. *If $K_j \to K$ pointwise and $|K_j| \leq M$, then* $\det(I + T\chi_B K_j) \to \det(I + T\chi_B K)$.

Appendix C: Results from Analytic Number Theory

Lemma 7.7. *For $\Re(s) > 1$, $-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$.*

Theorem 7.8 (Prime Number Theorem)**.** $\pi(x) \sim \frac{x}{\log x}$, *equivalently* $\psi(x) \sim x$.

Proposition 7.9 (Summation by Parts)**.** *If $F(x) = \sum_{n \leq x} f(n)$, then*

$$\sum_{a \leq n \leq b} f(n)g(n) = F(b)g(b) - F(a)g(a) - \int_a^b F(t)g'(t) \, \mathrm{d}t.$$